T3 group factors: general EFT1

One-loop renormalisable RGBeta structure prefactors across T3 models.

$$j = \frac{d-1}{2}, \qquad C_2(d) = \frac{d^2-1}{4}, \qquad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Coupling and interaction notation

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$
$T(d)$	$d(d^2-1)/12$
$C_{LLS_1S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$
R_W	$16\pi^2 \beta_{C_5} \supset R_W \lambda_5 C_{LLS_1S_2}$

Additional general-interaction couplings

Symbol	Field content	Stored coupling / contraction
$\lambda_{X_5}^{(1)}$	$(S_1^\dagger)^2\, S_2\, S_2^\dagger$	`lambdaT3GeneralQ000211I1`

λ_{X_6}	$(S_1^\dagger)^4$	`lambdaT3GeneralQ000400I1`
λ_{X_9}	$S_1\,(S_1^\dagger)^3$	`lambdaT3GeneralQ001300I1`
$\lambda_{X_{14}}$	$(H^\dagger)^2\,S_1\,S_2$	`lambdaT3GeneralQ021010I1`
$\lambda_{X_{18}}$	$H\,H^\dagger\,(S_1^\dagger)^2$	`lambdaT3GeneralQ110200I1`
λ_{X_1}	$(S_2^\dagger)^4$	`lambdaT3GeneralQ000004I1`
λ_{X_2}	$S_2\,(S_2^\dagger)^3$	`lambdaT3GeneralQ000013I1`
$\lambda_{X_7}^{(1)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ001102I1`
$\lambda_{X_{12}}$	$(H^\dagger)^2\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ020101I1`
$\lambda_{X_{16}}$	$H\,H^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ110002I1`
$\lambda_{X_{15}}$	$(H^\dagger)^2\,(S_1)^2$	`lambdaT3GeneralQ022000I1`
$\lambda_{X_{17}}^{(1)}$	$H\,H^\dagger\,S_1^\dagger\,S_2^\dagger$	`lambdaT3GeneralQ110101I1`
$\lambda_{X_{10}}$	$(H^\dagger)^2\,(S_2^\dagger)^2$	`lambdaT3GeneralQ020002I1`
$\lambda_{X_5}^{(2)}$	$(S_1^\dagger)^2\,S_2\,S_2^\dagger$	`lambdaT3GeneralQ000211I2`
$\lambda_{X_7}^{(2)}$	$S_1\,S_1^\dagger\,(S_2^\dagger)^2$	`lambdaT3GeneralQ001102I2`
$y_1^{\rm conj}$	$L\,F\,S_i^\dagger$	`y1Conj`
$y_2^{\rm conj}$	$L\,F\,S_i^\dagger$	`y2Conj`

Field representations

run	S_1	S_2	α
A	$(1,0)$	$(3,1)$	0
A	$(1,-1)$	$(3,0)$	-2
A	$(1,-2)$	$(3,-1)$	-4
B	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	1
B	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	-1
B	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	-3
C	$(2,\frac{1}{2})$	$(2,\frac{3}{2})$	1
C	$(2,-\frac{1}{2})$	$(2,\frac{1}{2})$	-1
C	$(2,-\frac{3}{2})$	$(2,-\frac{1}{2})$	-3
D	$(3,0)$	$(1,1)$	0
D	$(3,1)$	$(1,2)$	2
D	$(3,-1)$	$(1,0)$	-2
E	$(3,0)$	$(3,1)$	0
E	$(3,1)$	$(3,2)$	2
E	$(3,-1)$	$(3,0)$	-2
E	$(3,-2)$	$(3,-1)$	-4

Complete general-interaction RGE prefactors

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
g_2^3	$-\frac{17}{6}$	$-\frac{17}{6}$	$-\frac{17}{6}$

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
g_2^3	$-\frac{5}{2}$	$-\frac{5}{2}$	$-\frac{5}{2}$

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
g_2^3	$-\frac{11}{6}$	$-\frac{11}{6}$	$-\frac{11}{6}$

$\mathcal{S}$	E ($\alpha=-4$)
g_2^3	$-\frac{11}{6}$

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
g_3^3	-7	-7	-7

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
g_3^3	-7	-7	-7

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
g_3^3	-7	-7	-7

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
g_3^3	-7	-7	-7

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
g_3^3	-7	-7	-7

$\mathcal{S}$	E ($\alpha=-4$)
g_3^3	-7

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
g_1^3	$\frac{47}{6}$	$\frac{43}{6}$	$\frac{55}{6}$

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
g_1^3	$\frac{17}{2}$	$\frac{43}{6}$	$\frac{17}{2}$

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)

g_1^3	$\frac{17}{2}$	$\frac{43}{6}$	$\frac{17}{2}$

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
g_1^3	$\frac{43}{6}$	$\frac{55}{6}$	$\frac{47}{6}$

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
g_1^3	$\frac{47}{6}$	$\frac{71}{6}$	$\frac{47}{6}$

$\mathcal{S}$	E ($\alpha=-4$)
g_1^3	$\frac{71}{6}$

$\beta^{(1)}_{\lambda_{12}^{(1)}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^*+3(\lambda_{X_9})^*)$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda_{12}^{(1)}$	-12	-12	-12
$g_1^2\lambda_{12}^{(1)}$	-6	-6	-30
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	8	8	8
$\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	4	4	4
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	20	0
$\lambda_{X_7}^{(1)}(5(\lambda_{X_2})^*+4(\lambda_{X_7}^{(1)})^*)$	0	4	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
g_1^4	0	0	48
g_2^4	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^*+(\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^*+(\lambda_{11})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^*+5(\lambda_{X_9})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_2})^*+4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^*+(\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0

$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_5(\lambda_5)^*$	3	3	3
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda_{12}^{(1)}$	-9	-9	-9
$g_1^2\lambda_{12}^{(1)}$	-15	-3	-15
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	6	6	6
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	6	6	6
$\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda_{12}^{(A)}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_9(\lambda_9)^*$	0	24	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_5(\lambda_5)^*$	3	3	3
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda_{12}^{(1)}$	-9	-9	-9
$g_1^2\lambda_{12}^{(1)}$	-15	-3	-15
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	6	6	6
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	6	6	6

$\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda_{12}^{(A)}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_9(\lambda_9)^*$	0	24	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	9	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	20	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda_{12}^{(1)}$	-12	-12	-12
$g_1^2\lambda_{12}^{(1)}$	-6	-30	-6
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	8	8	8
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	12
$\lambda_{X_7}^{(1)}(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
g_1^4	0	48	0
g_2^4	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*)$	4	0	0

$\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	4	4	4
$\lambda_{X_7}^{(1)}(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	4
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	2	2	2
$\lambda_{X_9}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 3(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	2	0	0
$g_2^2\lambda_{12}^{(1)}$	-24	-24	-24
$g_1^2\lambda_{12}^{(1)}$	-6	-30	-6
${\lambda_{12}^{(1)}}^2$	4	4	4
$\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_{X_2}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(5(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	2
g_1^4	0	48	0
g_2^4	$\frac{63}{5}$	$\frac{63}{5}$	$\frac{63}{5}$
${\lambda_{12}^{(A)}}^2$	16	16	16
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(4(\lambda_{X_5}^{(1)})^* + 5(\lambda_{X_9})^*)$	4	0	0
$\lambda_{12}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_2})^* + 4(\lambda_{X_7}^{(1)})^*)$	0	0	0
$\lambda_{X_9}(5(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	4	0	0
$\lambda_{X_5}^{(2)}((\lambda_{X_5}^{(2)})^* + (\lambda_{X_9})^*)$	4	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
$\lambda_{S_2}^{(1)}(8\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
$\lambda_{12}^{(1)}(4\lambda_{S_1}^{(A)} + 4\lambda_{S_2}^{(A)})$	1	1	1
${\lambda_{12}^{(\times)}}^2$	4	4	4
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	4	4	4
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	4	4	4
$\lambda_{X_2}(5(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4
$\lambda_{X_7}^{(2)}((\lambda_{X_2})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_5(\lambda_5)^*$	2
$\lambda_{X_9}(\lambda^{(1)}_{X_5})^*$	0
$\lambda^{(1)}_{X_5}(4(\lambda^{(1)}_{X_5})^*+3(\lambda_{X_9})^*)$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_2^2\lambda^{(1)}_{12}$	-24
$g_1^2\lambda^{(1)}_{12}$	-30
${\lambda^{(1)}_{12}}^2$	4
$\lambda^{(1)}_{12}\lambda^{(1)}_{S_1}$	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{S_2}$	0
$\lambda^{(1)}_{12}\lambda^{(A)}_{S_2}$	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{HS_2}$	4
$\lambda_{X_2}(\lambda^{(1)}_{X_7})^*$	0
$\lambda^{(1)}_{X_7}(5(\lambda_{X_2})^*+4(\lambda^{(1)}_{X_7})^*)$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
g_1^4	48
g_2^4	$\frac{63}{5}$
${\lambda^{(A)}_{12}}^2$	16
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{(1)}_{X_{17}}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+2(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{10}((\lambda_{10})^*+(\lambda_{11})^*)$	0
$\lambda_{11}((\lambda_{10})^*+(\lambda_{11})^*)$	0
$\lambda^{(1)}_{X_5}(4(\lambda^{(1)}_{X_5})^*+5(\lambda_{X_9})^*)$	0
$\lambda^{(1)}_{12}\lambda^{(A)}_{S_1}$	0
$\lambda^{(1)}_{X_7}(3(\lambda_{X_2})^*+4(\lambda^{(1)}_{X_7})^*)$	0
$\lambda_{X_9}(5(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	0
$\lambda^{(2)}_{X_5}((\lambda^{(2)}_{X_5})^*+(\lambda_{X_9})^*)$	0
$\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	1
$\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	1
$\lambda^{(1)}_{12}(4\lambda^{(A)}_{S_1}+4\lambda^{(A)}_{S_2})$	1
${\lambda^{(\times)}_{12}}^2$	4
$\lambda^{(\times)}_{12}\lambda^{(A)}_{S_1}$	4
$\lambda^{(\times)}_{12}\lambda^{(A)}_{S_2}$	4
$\lambda_{X_2}(5(\lambda^{(1)}_{X_7})^*+(\lambda^{(2)}_{X_7})^*)$	0
$\lambda^{(2)}_{X_7}((\lambda_{X_2})^*+(\lambda^{(2)}_{X_7})^*)$	0

$\beta^{(1)}_{\lambda^{(A)}_{12}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_5(\lambda_5)^*$	0	0	0
$g_1^2(18g_2^2-15\lambda^{(A)}_{12})$	0	0	0
$g_2^2\lambda^{(A)}_{12}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(A)}_{12}$	0	0	0
$\lambda^{(A)}_{12}\lambda^{(1)}_{S_1}$	0	0	0
$\lambda^{(A)}_{12}\lambda^{(1)}_{S_2}$	0	0	0
$\lambda^{(A)}_{HS_1}\lambda^{(A)}_{HS_2}$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{(1)}_{X_{17}})^*$	0	0	0

$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_5(\lambda_5)^*$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	1	0	1
$g_2^2\lambda_{12}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	4	0
$\lambda_9(\lambda_9)^*$	0	-64	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	-4	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	-4	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	1	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_5(\lambda_5)^*$	$\frac{4}{3}$	$\frac{4}{3}$	$\frac{4}{3}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	1	0	1
$g_2^2\lambda_{12}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	4	0
$\lambda_9(\lambda_9)^*$	0	-64	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	-4	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	-4	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	1	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0

$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	0	0	0
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	0	0	0
$g_2^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{12}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	0	0	0
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	$-\frac{2}{9}$	$-\frac{2}{9}$	$-\frac{2}{9}$
$g_1^2(18g_2^2 - 15\lambda_{12}^{(A)})$	0	0	0
$g_2^2\lambda_{12}^{(A)}$	-24	-24	-24
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8	8	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	3	3	3
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{11}((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{12}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{1}{3}$	0	0
$g_1^2\lambda_{12}^{(A)}$	-6	0	-6
$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	$\frac{16}{3}$	$\frac{16}{3}$	$\frac{16}{3}$
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	16	16	16
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	16	16	16
$g_1^2(16g_2^2 - 30\lambda_{12}^{(A)})$	0	1	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{1}{3}$

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_5(\lambda_5)^*$	$-\frac{2}{9}$
$g_1^2(18g_2^2-15\lambda_{12}^{(A)})$	0
$g_2^2\lambda_{12}^{(A)}$	-24
$\lambda_{12}^{(1)}\lambda_{12}^{(A)}$	8
$\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	2
$\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	2
$\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$	3
$\lambda_{X_{17}}^{(1)}(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^*$	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_{X_{17}}^{(1)})^*$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{10}(5(\lambda_{10})^*+(\lambda_{11})^*)$	0
$\lambda_{11}((\lambda_{10})^*+5(\lambda_{11})^*)$	0
$g_1^2(-6g_2^2-3\lambda_{12}^{(A)})$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2\lambda_{12}^{(A)}$	0
$\lambda_{12}^{(A)}\lambda_{12}^{(\times)}$	$\frac{16}{3}$
$\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	16
$\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	16
$g_1^2(16g_2^2-30\lambda_{12}^{(A)})$	1
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_{12}^{(\times)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^*+8(\lambda_{X_7}^{(1)})^*+10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^*+10(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0

$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	0	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{12}^{(\times)}$	0	0	0
$g_1^2\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	0	0	0

$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_5(\lambda_5)^*$	-1	-1	-1
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	1	0	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	8	0	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	-1	0	0
g_2^4	$\frac{51}{10}$	$\frac{51}{10}$	$\frac{51}{10}$
$g_2^2\lambda_{12}^{(\times)}$	-24	-24	-24
$g_1^2\lambda_{12}^{(\times)}$	-6	-30	-6
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8	8	8
${\lambda_{12}^{(\times)}}^2$	10	10	10
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	-2	-2	-2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	-2	-2	-2
$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0	0	1
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0	0	4
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0	0	8
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	-1

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_5(\lambda_5)^*$	-1
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + 10(\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_9})^*)$	0
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(2)})^*$	0
$\lambda_{X_9}(\lambda_{X_5}^{(2)})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
g_2^4	$\frac{51}{10}$
$g_2^2\lambda_{12}^{(\times)}$	-24
$g_1^2\lambda_{12}^{(\times)}$	-30
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8
${\lambda_{12}^{(\times)}}^2$	10
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(1)}$	2
$\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	-2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(1)}$	2
$\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	-2
$\lambda_{X_7}^{(2)}(4(\lambda_{X_2})^* + 8(\lambda_{X_7}^{(1)})^* + 10(\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_2}(\lambda_{X_7}^{(2)})^*$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(2)})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_1}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_1(12\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 4\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 12\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1

$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	2	2	2
${\lambda_{HS_2}^{(1)}}^2$	6	6	6
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	24	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
${\lambda_{HS_1}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	4	0
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	4	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0

$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
${\lambda_{HS_1}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	4	0
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	4	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger)+4\,\text{Tr}(Y_e(Y_e)^\dagger)+12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	6	6	6
${\lambda_{HS_2}^{(1)}}^2$	2	2	2
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	8
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	3	3	3
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger)+4\,\text{Tr}(Y_e(Y_e)^\dagger)+12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$

$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1	1	1
$g_2^2\lambda_1$	-9	-9	-9
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_1}^2$	12	12	12
${\lambda_{HS_1}^{(1)}}^2$	6	6	6
${\lambda_{HS_2}^{(1)}}^2$	6	6	6
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
T	-4	-4	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	24
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	3	3	3
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_1(12\,\text{Tr}(Y_d(Y_d)^\dagger) + 4\,\text{Tr}(Y_e(Y_e)^\dagger) + 12\,\text{Tr}(Y_u(Y_u)^\dagger))$	1
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
g_2^4	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1
$g_2^2\lambda_1$	-9
g_1^4	$\frac{3}{4}$
${\lambda_1}^2$	12
${\lambda_{HS_1}^{(1)}}^2$	6
${\lambda_{HS_2}^{(1)}}^2$	6
${\lambda_{HS_2}^{(A)}}^2$	3
T	-4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
${\lambda_{HS_1}^{(A)}}^2$	3
$\lambda_7(\lambda_7)^*$	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda^{(1)}_{X_{17}}(\lambda^{(1)}_{X_{17}})^*$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0

$\beta^{(1)}_{\lambda^{(1)}_{HS_1}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda^{(1)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	12	12	12
$\lambda_{X_9}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	12	0	0

$g_2^2\lambda_{HS_1}^{(1)}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2\lambda_{HS_1}^{(1)}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{51}{2}$
$\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	6	6	6
$\lambda_1{\lambda_{HS_1}^{(1)}}^2$	6	6	6
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
$\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	4	4	4
$\lambda_{X_{16}}(\lambda_{X_7}^{(1)})^*$	0	12	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{16}})^*$	0	12	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	12	0
g_1^4	0	3	12
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HS_1}^{(1)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_{18}}(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{X_{16}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{HS_1}^{(1)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda_{HS_1}^{(1)}$	-9	-9	-9
$g_1^2\lambda_{HS_1}^{(1)}$	-3	-3	-15
$\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	4	4	4
$\lambda_1{\lambda_{HS_1}^{(1)}}^2$	6	6	6
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
$\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	6	6	6
$\lambda_{X_{16}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda_{HS_1}^{(A)}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_7(\lambda_7)^*$	0	24	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda_{X_{17}}^{(1)})^*)$	0	3	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{X_{17}}^{(1)}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0	1	0
$\lambda_{HS_1}^{(1)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_{18}}(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\lambda_{X_{16}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	-9	-9	-9
$g_1^2\lambda^{(1)}_{HS_1}$	-3	-3	-15
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	6	6	6
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{27}{4}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	24	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
${\lambda^{(A)}_{HS_1}}^2$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_7(\lambda_7)^*$	0	24	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-(\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+3(\lambda_{11})^*+2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-3(\lambda_{11})^*+4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{S_1})$	0	0	0
$\lambda_{X_{18}}(5(\lambda_{X_9})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12}+4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^*+(\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(1)}_{HS_1}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	20	0	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	2	2	2
$\lambda_1\lambda^{(1)}_{HS_1}$	6	6	6
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8	8	8
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	4
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	4
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
g_1^4	0	3	3
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	6	6	6
${\lambda^{(A)}_{HS_1}}^2$	6	6	6
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-(\lambda^{(1)}_{X_{17}})^*)$	0	0	0

$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	1	1	1
$\lambda_{X_{18}}(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	4	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	20	0	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{3}{2}$	$-\frac{15}{2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	0	0	0
$\lambda_1\lambda^{(1)}_{HS_1}{}^2$	6	6	6
$\lambda^{(1)}_{HS_1}$	4	4	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8	8	8
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda^{(1)}_{X_7}(\lambda_{X_{16}})^*$	0	0	12
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
g_1^4	0	3	3
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	6	6	6
$\lambda^{(A)\ 2}_{HS_1}$	6	6	6
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{11})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{11})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_1})$	1	1	1
$\lambda_{X_{18}}(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	4	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	1	1	1
$\lambda_{X_{16}}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	4
$\lambda^{(2)}_{X_7}(\lambda_{X_{16}})^*$	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{(1)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_9}(\lambda_{X_{18}})^*$	0
$\lambda_{X_{18}}(3(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_2^2\lambda^{(1)}_{HS_1}$	$-\frac{33}{2}$
$g_1^2\lambda^{(1)}_{HS_1}$	$-\frac{51}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_2}$	0
$\lambda_1\lambda^{(1)}_{HS_1}{}^2$	6
$\lambda^{(1)}_{HS_1}$	4
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{S_1}$	8
$\lambda_{X_{16}}(\lambda^{(1)}_{X_7})^*$	0

$\lambda_{X_7}^{(1)}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
g_1^4	12
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
g_2^4	6
$\lambda_{HS_1}^{(A)\ 2}$	6
$\lambda_7(\lambda_7)^*$	0
$\lambda_{11}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* - (\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(4(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + 3(\lambda_{11})^* + 2(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{X_{17}}^{(1)}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* - 3(\lambda_{11})^* + 4(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda_{HS_1}^{(1)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	1
$\lambda_{X_{18}}(5(\lambda_{X_9})^* + 4(\lambda_{X_{18}})^*)$	0
$\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1
$\lambda_{X_{16}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{16}})^*$	0

$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_1\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}})^* + (\lambda_{11})^*)$	0	0	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	$\frac{16}{3}$	$\frac{16}{3}$	$\frac{16}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	64	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	1	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1

$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	-64	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	4	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	1	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	1
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_5(\lambda_5)^*$	$\frac{16}{3}$	$\frac{16}{3}$	$\frac{16}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	64	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	1	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	-9	-9	-9
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	1	1	1
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	8	8	8
$\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	-64	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	4	0
$\lambda_{X_{17}}^{(1)}((\lambda_{11})^* - 2(\lambda_{X_{17}}^{(1)})^*)$	0	4	0
$g_1^2(-6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	1	0
$g_1^2(-18g_2^2 - 15\lambda_{HS_1}^{(A)})$	0	0	1
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2\lambda_{HS_1}^{(A)}$	0	0	0
$g_1^2(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2(-4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2})$	0	0	0
$g_1^2(-8g_2^2 - \frac{51\lambda_{HS_1}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{HS_1}^{(A)}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_5(\lambda_5)^*$	$-\frac{8}{3}$	$-\frac{8}{3}$	$-\frac{8}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2 - 3\lambda_{HS_1}^{(A)})$	0	0	0
$g_2^2\lambda_{HS_1}^{(A)}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$	0	0	0
$\lambda_1\lambda_{HS_1}^{(A)}$	2	2	2

$\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$	8	8	8
$\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0	0	0
$g_1^2(-18g_2^2 - 15\lambda^{(A)}_{HS_1})$	0	0	0
$\lambda^{(A)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda^{(A)}_{S_1})$	1	1	1
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$\frac{8}{3}$	0	0
$g_1^2\lambda^{(A)}_{HS_1}$	$-\frac{3}{2}$	0	0
$g_1^2(4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0	1	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{8}{3}$
$g_1^2(-4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0	0	1
$g_1^2(-8g_2^2 - \frac{51\lambda^{(A)}_{HS_1}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(A)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_5(\lambda_5)^*$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
$g_1^2(6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0	0	0
$g_2^2\lambda^{(A)}_{HS_1}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}$	12	12	12
$\lambda_1\lambda^{(A)}_{HS_1}$	2	2	2
$\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$	8	8	8
$\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$	2	2	2
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$g_1^2(-6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0	0	0
$g_1^2(-18g_2^2 - 15\lambda^{(A)}_{HS_1})$	0	0	0
$\lambda^{(A)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda^{(A)}_{S_1})$	1	1	1
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{8}{9}$	0	0
$g_1^2\lambda^{(A)}_{HS_1}$	$-\frac{3}{2}$	0	0
$g_1^2(4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0	1	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{4}{3}$
$g_1^2(-4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0	0	1
$g_1^2(-8g_2^2 - \frac{51\lambda^{(A)}_{HS_1}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{(A)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_5(\lambda_5)^*$	$-\frac{4}{3}$
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
$g_1^2(6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0
$g_2^2\lambda^{(A)}_{HS_1}$	$-\frac{33}{2}$
$\lambda^{(A)}_{12}\lambda^{(A)}_{HS_2}$	12
$\lambda_1\lambda^{(A)}_{HS_1}$	2
$\lambda^{(1)}_{HS_1}\lambda^{(A)}_{HS_1}$	8

$\lambda^{(A)}_{HS_1}\lambda^{(1)}_{S_1}$	2
$\lambda_7(\lambda_7)^*$	0
$\lambda_{11}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + (\lambda_{11})^*)$	0
$\lambda^{(1)}_{X_{17}}((\lambda_{11})^* - 2(\lambda^{(1)}_{X_{17}})^*)$	0
$g_1^2(-6g_2^2 - 3\lambda^{(A)}_{HS_1})$	0
$g_1^2(-18g_2^2 - 15\lambda^{(A)}_{HS_1})$	0
$\lambda^{(A)}_{HS_1}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 16\lambda^{(A)}_{S_1})$	1
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2\lambda^{(A)}_{HS_1}$	0
$g_1^2(4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$g_1^2(-4g_2^2 - \frac{15\lambda^{(A)}_{HS_1}}{2})$	0
$g_1^2(-8g_2^2 - \frac{51\lambda^{(A)}_{HS_1}}{2})$	1

$\beta^{(1)}_{\lambda^{(1)}_{HS_2}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda^{(1)}_{HS_2}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	4	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	4	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
g_2^4	6	6	6
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
g_1^4	3	0	3
$g_1^2\lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$	$-\frac{3}{2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	2	2	2
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}{}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	8	8	8
$\lambda^{(A)}_{HS_2}$	6	6	6
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	20	0
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	4	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	4	0
$\lambda^{(1)}_{HS_2}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda^{(1)}_{HS_2}(6\,\text{Tr}(Y_d(Y_d)^\dagger) + 2\,\text{Tr}(Y_e(Y_e)^\dagger) + 6\,\text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_5(\lambda_5)^*$	6	6	6

$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_2^2\lambda^{(1)}_{HS_2}$	-9	-9	-9
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_1^2\lambda^{(1)}_{HS_2}$	-15	-3	-3
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}{}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	6	6	6
$\lambda^{(A)}_{HS_2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda_8(\lambda_8)^*$	0	24	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	24
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_5(\lambda_5)^*$	6	6	6
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_2^2\lambda^{(1)}_{HS_2}$	-9	-9	-9
g_1^4	$\frac{27}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$g_1^2\lambda^{(1)}_{HS_2}$	-15	-3	-3
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	4	4	4
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}{}^2$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	6	6	6
$\lambda^{(A)}_{HS_2}$	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	3	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$\lambda_8(\lambda_8)^*$	0	24	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	24
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0

$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_5(\lambda_5)^*$	12	12	12
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	12	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	12	0	0
g_2^4	0	0	0
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
g_1^4	3	12	0
$g_1^2\lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{3}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	6	6	6
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	4	4	4
$\lambda^{(A)}_{HS_2}$	0	0	0
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	12
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	12
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	4
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger) + 4\lambda^{(A)}_{S_2})$	1	1	1
$\lambda_5(\lambda_5)^*$	4	4	4
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	12	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	4	0	0
g_2^4	6	6	6
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
g_1^4	3	12	0
$g_1^2\lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$	$-\frac{51}{2}$	$-\frac{3}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	0	0	0
$\lambda_1\lambda^{(1)}_{HS_2}$	6	6	6
$\lambda^{(1)}_{HS_2}$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	8	8	8
$\lambda^{(A)}_{HS_2}$	6	6	6
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0	0	20
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	4

$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	4
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0	0	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	4	0	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	4	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{S_2})$	1
$\lambda_5(\lambda_5)^*$	4
$\lambda_{X_{18}}(\lambda^{(1)}_{X_5})^*$	0
$\lambda^{(1)}_{X_5}(\lambda_{X_{18}})^*$	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
g_2^4	6
$g_2^2\lambda^{(1)}_{HS_2}$	$-\frac{33}{2}$
g_1^4	3
$g_1^2\lambda^{(1)}_{HS_2}$	$-\frac{15}{2}$
$\lambda^{(1)}_{12}\lambda^{(1)}_{HS_1}$	0
$\lambda_1\lambda^{(1)}_{HS_2}$	6
$\lambda^{(1)}_{HS_2}{}^2$	4
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{S_2}$	8
$\lambda^{(A)}_{HS_2}{}^2$	6
$\lambda_{X_2}(\lambda_{X_{16}})^*$	0
$\lambda_{X_{16}}(5(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$\lambda^{(1)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - (\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* + 3(\lambda_{10})^* + 2(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda^{(1)}_{X_{17}}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^* - 3(\lambda_{10})^* + 4(\lambda^{(1)}_{X_{17}})^*)$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$\lambda_{X_{16}}(3(\lambda_{X_2})^* + 4(\lambda_{X_{16}})^*)$	0
$\lambda_{X_{18}}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0
$\lambda^{(2)}_{X_5}(\lambda_{X_{18}})^*$	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{12} + 4\lambda^{(\times)}_{12})$	1

$\beta^{(1)}_{\lambda^{(A)}_{HS_2}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda^{(A)}_{S_2})$	1	1	1
$\lambda_5(\lambda_5)^*$	$\frac{8}{3}$	$\frac{8}{3}$	$\frac{8}{3}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$\frac{8}{3}$	0	0
$g_1^2(4g_2^2 - \frac{15\lambda^{(A)}_{HS_2}}{2})$	1	0	0
$g_2^2\lambda^{(A)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$

$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	$-\frac{8}{3}$	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	$-\frac{3}{2}$	0
$g_1^2(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	1
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2-15\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	0	0	0
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*+(\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^*-4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2-3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2-3\lambda_{HS_2}^{(A)})$	0	0	0
$g_1^2(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_5(\lambda_5)^*$	16	16	16
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	-9	-9	-9
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8	8	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	0
$g_1^2(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$g_1^2(18g_2^2-15\lambda_{HS_2}^{(A)})$	1	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	1	1	1
$\lambda_{10}((\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	-4	0
$\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}}(2(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\mathrm{Cross}})^*+(\lambda_{10})^*)$	0	-4	0
$\lambda_8(\lambda_8)^*$	0	64	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^*-4(\lambda_{10})^*)$	0	1	0
$g_1^2(6g_2^2-3\lambda_{HS_2}^{(A)})$	0	1	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	-64
$g_1^2(-6g_2^2-3\lambda_{HS_2}^{(A)})$	0	0	1
$g_1^2(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_5(\lambda_5)^*$	16	16	16
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$g_2^2\lambda_{HS_2}^{(A)}$	-9	-9	-9
$\lambda_1\lambda_{HS_2}^{(A)}$	2	2	2

$\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	8	8	8
$\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$	2	2	2
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda^{(A)}_{HS_2}$	0	0	0
$g_1^2(-4g_2^2-\frac{15\lambda^{(A)}_{HS_2}}{2})$	0	0	0
$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	1	1
$g_1^2(18g_2^2-15\lambda^{(A)}_{HS_2})$	1	0	0
$\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	1	1	1
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda^{(1)}_{X_{17}})^*)$	0	-4	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{10})^*)$	0	-4	0
$\lambda_8(\lambda_8)^*$	0	64	0
$\lambda^{(1)}_{X_{17}}(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda_{10})^*)$	0	1	0
$g_1^2(6g_2^2-3\lambda^{(A)}_{HS_2})$	0	1	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	-64
$g_1^2(-6g_2^2-3\lambda^{(A)}_{HS_2})$	0	0	1
$g_1^2(8g_2^2-\frac{51\lambda^{(A)}_{HS_2}}{2})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda^{(A)}_{S_2})$	0	0	0
$\lambda_5(\lambda_5)^*$	0	0	0
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0	0	0
$g_1^2(4g_2^2-\frac{15\lambda^{(A)}_{HS_2}}{2})$	0	0	0
$g_2^2\lambda^{(A)}_{HS_2}$	0	0	0
$\lambda_1\lambda^{(A)}_{HS_2}$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	0	0	0
$\lambda^{(A)}_{HS_2}\lambda^{(1)}_{S_2}$	0	0	0
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	0
$g_1^2\lambda^{(A)}_{HS_2}$	0	0	0
$g_1^2(-4g_2^2-\frac{15\lambda^{(A)}_{HS_2}}{2})$	0	0	0
$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2-15\lambda^{(A)}_{HS_2})$	0	0	0
$\lambda^{(A)}_{12}\lambda^{(A)}_{HS_1}$	0	0	0
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda^{(1)}_{X_{17}}(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2-3\lambda^{(A)}_{HS_2})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2-3\lambda^{(A)}_{HS_2})$	0	0	0
$g_1^2(8g_2^2-\frac{51\lambda^{(A)}_{HS_2}}{2})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda^{(A)}_{HS_2}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda^{(A)}_{S_2})$	1	1	1
$\lambda_5(\lambda_5)^*$	$-\frac{8}{9}$	$-\frac{8}{9}$	$-\frac{8}{9}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	$-\frac{8}{9}$	0	0
$g_1^2(4g_2^2-\frac{15\lambda^{(A)}_{HS_2}}{2})$	1	0	0
$g_2^2\lambda^{(A)}_{HS_2}$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$\lambda_1\lambda^{(A)}_{HS_2}$	2	2	2
$\lambda^{(1)}_{HS_2}\lambda^{(A)}_{HS_2}$	8	8	8

$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2	2	2
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0	0	$-\frac{4}{3}$
$g_1^2\lambda_{HS_2}^{(A)}$	0	0	$-\frac{3}{2}$
$g_1^2(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0	0	0
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_1^2(18g_2^2-15\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	12	12	12
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0	0	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{10})^*)$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^*-4(\lambda_{10})^*)$	0	0	0
$g_1^2(6g_2^2-3\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2(-6g_2^2-3\lambda_{HS_2}^{(A)})$	0	0	0
$g_1^2(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2})$	0	1	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+16\lambda_{S_2}^{(A)})$	1
$\lambda_5(\lambda_5)^*$	$-\frac{8}{9}$
$\lambda_{X_{14}}(\lambda_{X_{14}})^*$	0
$g_1^2(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	0
$g_2^2\lambda_{HS_2}^{(A)}$	$-\frac{33}{2}$
$\lambda_1\lambda_{HS_2}^{(A)}$	2
$\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	8
$\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	2
$\lambda_{X_{12}}(\lambda_{X_{12}})^*$	0
$g_1^2\lambda_{HS_2}^{(A)}$	0
$g_1^2(-4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2})$	1
$\lambda_{HS_2}^{(A)}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_1^2(18g_2^2-15\lambda_{HS_2}^{(A)})$	0
$\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$	12
$\lambda_{10}((\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{X_{17}}^{(1)})^*)$	0
$\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(2(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*+(\lambda_{10})^*)$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_{X_{17}}^{(1)}(8(\lambda_{X_{17}}^{(1)})^*-4(\lambda_{10})^*)$	0
$g_1^2(6g_2^2-3\lambda_{HS_2}^{(A)})$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$g_1^2(-6g_2^2-3\lambda_{HS_2}^{(A)})$	0
$g_1^2(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2})$	0

$\beta_{\lambda_7}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0

$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$g_2^2\lambda_7$	0	-9	0
$g_1^2\lambda_7$	0	-3	0
$\lambda_1\lambda_7$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	8	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	-4	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	2	0
$\lambda_8\lambda_9$	0	4	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$g_2^2\lambda_7$	0	-9	0
$g_1^2\lambda_7$	0	-3	0
$\lambda_1\lambda_7$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	8	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	-4	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	2	0
$\lambda_8\lambda_9$	0	4	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_7$	0	0	0
$g_1^2\lambda_7$	0	0	0
$\lambda_1\lambda_7$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_7$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_7$	0	0	0
$\lambda_7\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_8\lambda_9$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_7(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_2^2\lambda_7$	0
$g_1^2\lambda_7$	0
$\lambda_1\lambda_7$	0

$\lambda^{(1)}_{HS_1}\lambda_7$	0
$\lambda^{(A)}_{HS_1}\lambda_7$	0
$\lambda_7\lambda^{(1)}_{S_1}$	0
$\lambda_8\lambda_9$	0

$\beta^{(1)}_{\lambda_{12}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0	0	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	-9	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	-3	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	2	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0	1	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0	1	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0	1	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0	1	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0	1	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0	1	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0	1	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	-9	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	-3	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	2	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0	1	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0	1	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0	1	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0	1	0

$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0	1	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0	1	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0	0	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0	0	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0	0	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}(6\,\text{Tr}(Y_d(Y_d)^\dagger)+2\,\text{Tr}(Y_e(Y_e)^\dagger)+6\,\text{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_9(8(\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*-4(\lambda^{(1)}_{X_{17}})^*)$	0
$g_2^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$g_1^2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$\lambda^{(1)}_{12}\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}$	0
$\lambda^{(A)}_{12}(\lambda^{(1)}_{X_{17}}-\frac{\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}}{2})$	0
$\lambda_1(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-2\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_1}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{11})$	0
$\lambda^{(A)}_{HS_1}(2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}+\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_2}(4\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+6\lambda_{10})$	0
$\lambda^{(A)}_{HS_2}(-2\lambda^{\text{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}-\lambda^{(1)}_{X_{17}})$	0

$\beta^{(1)}_{\lambda_8}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	0	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	0	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_7(\lambda_9)^*$	0	4	0
$g_2^2\lambda_8$	0	-9	0
$g_1^2\lambda_8$	0	-3	0
$\lambda_1\lambda_8$	0	2	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	8	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	4	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	2	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_7(\lambda_9)^*$	0	4	0
$g_2^2\lambda_8$	0	-9	0
$g_1^2\lambda_8$	0	-3	0
$\lambda_1\lambda_8$	0	2	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	8	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	4	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	2	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda_8$	0	0	0
$\lambda^{(A)}_{HS_2}\lambda_8$	0	0	0
$\lambda_8\lambda^{(1)}_{S_2}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_7(\lambda_9)^*$	0	0	0
$g_2^2\lambda_8$	0	0	0
$g_1^2\lambda_8$	0	0	0
$\lambda_1\lambda_8$	0	0	0

$\lambda_{HS_2}^{(1)}\lambda_8$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0	0	0
$\lambda_8\lambda_{S_2}^{(1)}$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_8(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_7(\lambda_9)^*$	0
$g_2^2\lambda_8$	0
$g_1^2\lambda_8$	0
$\lambda_1\lambda_8$	0
$\lambda_{HS_2}^{(1)}\lambda_8$	0
$\lambda_{HS_2}^{(A)}\lambda_8$	0
$\lambda_8\lambda_{S_2}^{(1)}$	0

$\beta^{(1)}_{\lambda_{S_1}^{(1)}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	288	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	108	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	8	0	0
${\lambda_{12}^{(1)}}^2$	6	6	6
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{S_1}^{(1)}}^2$	10	10	10
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	24	0
g_1^4	0	12	192
$g_1^2\lambda_{S_1}^{(1)}$	0	-12	-48
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	0	0	0
$g_1^2(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2}-27\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$g_1^2(8g_2^2-12\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3}+4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^*+\frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2-48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
${\lambda^{(1)}_{12}}^2$	4	4	4
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
${\lambda^{(1)}_{S_1}}^2$	12	12	12
$\lambda^{(1)}_{X_7}(\lambda^{(1)}_{X_7})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{243}{4}$
$g_1^2\lambda^{(1)}_{S_1}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda^{(1)}_{S_1})$	1	1	0
$g_2^2\lambda^{(1)}_{S_1}$	-9	-9	-9
${\lambda^{(A)}_{12}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda^{(A)}_{HS_1}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{11}(\lambda_{11})^*$	0	24	0
$g_1^2(\frac{27g_2^2}{2}-27\lambda^{(1)}_{S_1})$	0	0	1
$\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$	0	0	0
${\lambda^{(A)}_{S_1}}^2$	0	0	0
$g_1^2(8g_2^2-12\lambda^{(1)}_{S_1})$	0	0	0
$\lambda^{(1)}_{X_5}(3(\lambda^{(1)}_{X_5})^*+(\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(2)}_{X_5}(\frac{8(\lambda^{(1)}_{X_5})^*}{3}+4(\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$	0	0	0
${\lambda^{(\times)}_{12}}^2$	0	0	0
$\lambda^{(1)}_{X_7}(3(\lambda^{(1)}_{X_7})^*+(\lambda^{(2)}_{X_7})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(8(\lambda^{(1)}_{X_7})^*+\frac{20(\lambda^{(2)}_{X_7})^*}{3})$	0	0	0
$g_1^2(32g_2^2-48\lambda^{(1)}_{S_1})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
${\lambda^{(1)}_{12}}^2$	4	4	4
${\lambda^{(1)}_{HS_1}}^2$	4	4	4
${\lambda^{(1)}_{S_1}}^2$	12	12	12
$\lambda^{(1)}_{X_7}(\lambda^{(1)}_{X_7})^*$	0	0	0
g_1^4	$\frac{3}{4}$	$\frac{3}{4}$	$\frac{243}{4}$
$g_1^2\lambda^{(1)}_{S_1}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	8	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2}-3\lambda^{(1)}_{S_1})$	1	1	0
$g_2^2\lambda^{(1)}_{S_1}$	-9	-9	-9
${\lambda^{(A)}_{12}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda^{(A)}_{HS_1}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$

$\lambda_7(\lambda_7)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{11}(\lambda_{11})^*$	0	24	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	1
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	$\frac{8}{3}$	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$\frac{832}{3}$	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	$\frac{352}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$\frac{16}{3}$	0	0
${\lambda_{12}^{(1)}}^2$	2	2	2
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{S_1}^{(1)}}^2$	14	14	14
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	8
g_1^4	0	12	12
$g_1^2\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	-24	-24	-24
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	8	8	8
${\lambda_{S_1}^{(A)}}^2$	16	16	16
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	1	1
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)

$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$\frac{832}{3}$	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	$\frac{352}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$\frac{16}{3}$	0	0
${\lambda_{12}^{(1)}}^2$	6	6	6
${\lambda_{HS_1}^{(1)}}^2$	4	4	4
${\lambda_{S_1}^{(1)}}^2$	14	14	14
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
g_1^4	0	12	12
$g_1^2\lambda_{S_1}^{(1)}$	0	0	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0	0	0
$g_2^2\lambda_{S_1}^{(1)}$	-24	-24	-24
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_7(\lambda_7)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{11}(\lambda_{11})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	8	8	8
${\lambda_{S_1}^{(A)}}^2$	16	16	16
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0	1	1
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	$\frac{8}{3}$	0	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	1	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8	8	8
${\lambda_{12}^{(\times)}}^2$	$\frac{20}{3}$	$\frac{20}{3}$	$\frac{20}{3}$
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	8
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0	0	1
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
${\lambda_{12}^{(1)}}^2$	6
${\lambda_{HS_1}^{(1)}}^2$	4
${\lambda_{S_1}^{(1)}}^2$	14
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0
g_1^4	192
$g_1^2\lambda_{S_1}^{(1)}$	0
$\lambda_{X_{15}}(\lambda_{X_{15}})^*$	0
g_2^4	22
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_1}^{(1)})$	0
$g_2^2\lambda_{S_1}^{(1)}$	-24
${\lambda_{12}^{(A)}}^2$	0
${\lambda_{HS_1}^{(A)}}^2$	0
$\lambda_7(\lambda_7)^*$	0
$\lambda_9(\lambda_9)^*$	0

$\lambda_{11}(\lambda_{11})^*$	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_1}^{(1)})$	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	8
${\lambda_{S_1}^{(A)}}^2$	16
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(1)})$	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(\frac{8(\lambda_{X_5}^{(1)})^*}{3} + 4(\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	8
${\lambda_{12}^{(\times)}}^2$	$\frac{20}{3}$
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(8(\lambda_{X_7}^{(1)})^* + \frac{20(\lambda_{X_7}^{(2)})^*}{3})$	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(1)})$	1

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{HS_1}^{(A)}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	0	0	0
${\lambda_{S_1}^{(A)}}^2$	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0

$\lambda^{(2)}_{X_7}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda^{(A)}_{S_1})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0	0	0
g_2^4	0	0	0
$g_2^2\lambda^{(A)}_{S_1}$	0	0	0
${\lambda^{(A)}_{HS_1}}^2$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$	0	0	0
${\lambda^{(A)}_{S_1}}^2$	0	0	0
$g_1^2(8g_2^2 - 12\lambda^{(A)}_{S_1})$	0	0	0
$\lambda^{(2)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
${\lambda^{(A)}_{12}}^2$	0	0	0
${\lambda^{(\times)}_{12}}^2$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda^{(A)}_{S_1})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	$-\frac{4}{3}$	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$-\frac{224}{3}$	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	$-\frac{44}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$-\frac{8}{3}$	0	0
g_2^4	-2	-2	-2
$g_2^2\lambda^{(A)}_{S_1}$	-24	-24	-24
${\lambda^{(A)}_{HS_1}}^2$	3	3	3
$\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$	12	12	12
${\lambda^{(A)}_{S_1}}^2$	20	20	20
$g_1^2(8g_2^2 - 12\lambda^{(A)}_{S_1})$	0	1	1
$\lambda^{(2)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda^{(1)}_{X_5}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
${\lambda^{(A)}_{12}}^2$	0	0	0
${\lambda^{(\times)}_{12}}^2$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_1^2(32g_2^2 - 48\lambda^{(A)}_{S_1})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_6})^*$	$-\frac{224}{3}$	0	0
$\lambda_{X_9}(\lambda_{X_9})^*$	$-\frac{44}{3}$	0	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	$-\frac{8}{3}$	0	0
g_2^4	-2	-2	-2
$g_2^2\lambda^{(A)}_{S_1}$	-24	-24	-24
${\lambda^{(A)}_{HS_1}}^2$	3	3	3
$\lambda^{(1)}_{S_1}\lambda^{(A)}_{S_1}$	12	12	12

$\lambda_{S_1}^{(A)^2}$	20	20	20
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0	1	1
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	$-\frac{4}{3}$	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	$-\frac{4}{3}$	0	0
$\lambda_{12}^{(A)^2}$	12	12	12
$\lambda_{12}^{(\times)^2}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0	0	$-\frac{4}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_6}(\lambda_{X_6})^*$	0
$\lambda_{X_9}(\lambda_{X_9})^*$	0
$\lambda_{X_{18}}(\lambda_{X_{18}})^*$	0
g_2^4	-2
$g_2^2\lambda_{S_1}^{(A)}$	-24
$\lambda_{HS_1}^{(A)^2}$	3
$\lambda_{S_1}^{(1)}\lambda_{S_1}^{(A)}$	12
$\lambda_{S_1}^{(A)^2}$	20
$g_1^2(8g_2^2 - 12\lambda_{S_1}^{(A)})$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{12}^{(A)^2}$	12
$\lambda_{12}^{(\times)^2}$	$-\frac{4}{3}$
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(2)})^*$	0
$g_1^2(32g_2^2 - 48\lambda_{S_1}^{(A)})$	1

$\beta_{\lambda_{11}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_9((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2} - \frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} + \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_9((\lambda_{10})^* + 5(\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{11}$	0	-9	0
$g_1^2\lambda_{11}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2} - \frac{5\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} + \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0

$\lambda_{S_1}^{(1)}\lambda_{11}$	0	12	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{11}$	0	-9	0
$g_1^2\lambda_{11}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	12	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{11}$	0	0	0
$g_1^2\lambda_{11}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_9((\lambda_{10})^*+5(\lambda_{11})^*)$	0
$g_2^2\lambda_{11}$	0
$g_1^2\lambda_{11}$	0
$\lambda_{12}^{(1)}(6\lambda_{10}+6\lambda_{11})$	0
$\lambda_{12}^{(A)}(-\frac{\lambda_{10}}{2}-\frac{5\lambda_{11}}{2})$	0
$\lambda_{HS_1}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0
$\lambda_{HS_1}^{(A)}(\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}+\frac{\lambda_{X_{17}}^{(1)}}{2})$	0
$\lambda_{S_1}^{(1)}\lambda_{11}$	0

$\beta_{\lambda_9}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_7(\lambda_8)^*$	0	4	0
$g_2^2\lambda_9$	0	-9	0
$g_1^2\lambda_9$	0	-3	0
$\lambda_{12}^{(1)}\lambda_9$	0	8	0
$\lambda_{12}^{(A)}\lambda_9$	0	-4	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	2	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	2	0
${\lambda_{10}}^2$	0	5	0
$\lambda_{10}\lambda_{11}$	0	2	0
${\lambda_{11}}^2$	0	5	0
${\lambda_{X_{17}}^{(1)}}^2$	0	1	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_7(\lambda_8)^*$	0	4	0
$g_2^2\lambda_9$	0	-9	0
$g_1^2\lambda_9$	0	-3	0
$\lambda_{12}^{(1)}\lambda_9$	0	8	0
$\lambda_{12}^{(A)}\lambda_9$	0	-4	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0	1	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	2	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	2	0
${\lambda_{10}}^2$	0	5	0
$\lambda_{10}\lambda_{11}$	0	2	0
${\lambda_{11}}^2$	0	5	0
${\lambda_{X_{17}}^{(1)}}^2$	0	1	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0	0	0

$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_7(\lambda_8)^*$	0	0	0
$g_2^2\lambda_9$	0	0	0
$g_1^2\lambda_9$	0	0	0
$\lambda_{12}^{(1)}\lambda_9$	0	0	0
$\lambda_{12}^{(A)}\lambda_9$	0	0	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_9$	0	0	0
$\lambda_9\lambda_{S_2}^{(1)}$	0	0	0
${\lambda_{10}}^2$	0	0	0
$\lambda_{10}\lambda_{11}$	0	0	0
${\lambda_{11}}^2$	0	0	0
${\lambda_{X_{17}}^{(1)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_7(\lambda_8)^*$	0
$g_2^2\lambda_9$	0
$g_1^2\lambda_9$	0
$\lambda_{12}^{(1)}\lambda_9$	0
$\lambda_{12}^{(A)}\lambda_9$	0
${\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}^2$	0
$\lambda_{S_1}^{(1)}\lambda_9$	0
$\lambda_9\lambda_{S_2}^{(1)}$	0
${\lambda_{10}}^2$	0
$\lambda_{10}\lambda_{11}$	0
${\lambda_{11}}^2$	0
${\lambda_{X_{17}}^{(1)}}^2$	0

$\beta_{\lambda_{10}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{10}$	0	-9	0
$g_1^2\lambda_{10}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	12	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	4	0
$g_2^2\lambda_{10}$	0	-9	0
$g_1^2\lambda_{10}$	0	-3	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	1	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	1	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	1	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	1	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	12	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0	0	0
$g_2^2\lambda_{10}$	0	0	0
$g_1^2\lambda_{10}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0	0	0
$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2} - \frac{\lambda_{11}}{2})$	0	0	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} - 2\lambda_{X_{17}}^{(1)})$	0	0	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{\lambda_{X_{17}}^{(1)}}{2})$	0	0	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_9(5(\lambda_{10})^* + (\lambda_{11})^*)$	0
$g_2^2\lambda_{10}$	0
$g_1^2\lambda_{10}$	0
$\lambda_{12}^{(1)}(6\lambda_{10} + 6\lambda_{11})$	0

$\lambda_{12}^{(A)}(-\frac{5\lambda_{10}}{2}-\frac{\lambda_{11}}{2})$	0
$\lambda_{HS_2}^{(1)}(2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}-2\lambda_{X_{17}}^{(1)})$	0
$\lambda_{HS_2}^{(A)}(-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{X_{17}}^{(1)}}{2})$	0
$\lambda_{10}\lambda_{S_2}^{(1)}$	0

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	8	0	0
g_2^4	22	22	22
$g_1^2(8g_2^2-12\lambda_{S_2}^{(1)})$	1	0	1
$g_2^2\lambda_{S_2}^{(1)}$	-24	-24	-24
g_1^4	12	0	12
${\lambda_{12}^{(1)}}^2$	2	2	2
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	14	14	14
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	8	8	8
${\lambda_{S_2}^{(A)}}^2$	16	16	16
$\lambda_{X_1}(\lambda_{X_1})^*$	0	$\frac{832}{3}$	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	$\frac{352}{3}$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	$\frac{8}{3}$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	$\frac{16}{3}$	0
$g_1^2(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)})$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^*+\frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2-48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3}+4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(8g_2^2-12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-9	-9	-9
g_1^4	$\frac{243}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_{12}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	12	12	12

$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	1	0	0
${\lambda_{12}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{10}(\lambda_{10})^*$	0	24	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	1	1
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	$\frac{9}{4}$	$\frac{9}{4}$	$\frac{9}{4}$
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-9	-9	-9
g_1^4	$\frac{243}{4}$	$\frac{3}{4}$	$\frac{3}{4}$
${\lambda_{12}^{(1)}}^2$	4	4	4
${\lambda_{HS_2}^{(1)}}^2$	4	4	4
${\lambda_{S_2}^{(1)}}^2$	12	12	12
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	1	0	0
${\lambda_{12}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
${\lambda_{HS_2}^{(A)}}^2$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$
$\lambda_8(\lambda_8)^*$	0	8	0
$\lambda_9(\lambda_9)^*$	0	8	0
$\lambda_{10}(\lambda_{10})^*$	0	24	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	1	1
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	8
$g_1^2\lambda_{S_2}^{(1)}$	0	0	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0

$\lambda_{12}^{(\times)^2}$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	24	0	0
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	0	0	0
$g_2^2\lambda_{S_2}^{(1)}$	0	0	0
g_1^4	12	192	0
$\lambda_{12}^{(1)^2}$	6	6	6
$\lambda_{HS_2}^{(1)^2}$	4	4	4
$\lambda_{S_2}^{(1)^2}$	10	10	10
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{S_2}^{(A)^2}$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	288
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	108
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	12
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	8
$g_1^2(\frac{27g_2^2}{2} - 27\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{12}^{(A)^2}$	0	0	0
$\lambda_{HS_2}^{(A)^2}$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda_{S_2}^{(1)}$	-12	-48	0
$\lambda_{X_5}^{(1)}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(8(\lambda_{X_5}^{(1)})^* + \frac{20(\lambda_{X_5}^{(2)})^*}{3})$	0	0	0
$\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(1)})$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_5}^{(1)}(\lambda_{X_5}^{(1)})^*$	0	0	0
g_2^4	22	22	22
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(1)})$	1	0	0
$g_2^2\lambda_{S_2}^{(1)}$	-24	-24	-24
g_1^4	12	192	0
$\lambda_{12}^{(1)^2}$	6	6	6
$\lambda_{HS_2}^{(1)^2}$	4	4	4
$\lambda_{S_2}^{(1)^2}$	14	14	14
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	8	8	8
$\lambda_{S_2}^{(A)^2}$	16	16	16

$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	$\frac{832}{3}$
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	$\frac{352}{3}$
$\lambda^{(1)}_{X_7}(\lambda^{(1)}_{X_7})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	$\frac{16}{3}$
$g_1^2(\frac{27g_2^2}{2} - 27\lambda^{(1)}_{S_2})$	0	0	0
${\lambda^{(A)}_{12}}^2$	0	0	0
${\lambda^{(A)}_{HS_2}}^2$	0	0	0
$\lambda_8(\lambda_8)^*$	0	0	0
$\lambda_9(\lambda_9)^*$	0	0	0
$\lambda_{10}(\lambda_{10})^*$	0	0	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_2})$	0	0	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0	0	0
$g_1^2\lambda^{(1)}_{S_2}$	0	0	0
$\lambda^{(1)}_{X_5}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	8	0	0
$\lambda^{(2)}_{X_5}(8(\lambda^{(1)}_{X_5})^* + \frac{20(\lambda^{(2)}_{X_5})^*}{3})$	1	0	0
$\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$	8	8	8
${\lambda^{(\times)}_{12}}^2$	$\frac{20}{3}$	$\frac{20}{3}$	$\frac{20}{3}$
$g_1^2(32g_2^2 - 48\lambda^{(1)}_{S_2})$	0	1	0
$\lambda^{(1)}_{X_7}(3(\lambda^{(1)}_{X_7})^* + (\lambda^{(2)}_{X_7})^*)$	0	0	$\frac{8}{3}$
$\lambda^{(2)}_{X_7}(\frac{8(\lambda^{(1)}_{X_7})^*}{3} + 4(\lambda^{(2)}_{X_7})^*)$	0	0	1

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda^{(1)}_{X_5}(\lambda^{(1)}_{X_5})^*$	0
g_2^4	22
$g_1^2(8g_2^2 - 12\lambda^{(1)}_{S_2})$	1
$g_2^2\lambda^{(1)}_{S_2}$	-24
g_1^4	12
${\lambda^{(1)}_{12}}^2$	6
${\lambda^{(1)}_{HS_2}}^2$	4
${\lambda^{(1)}_{S_2}}^2$	14
$\lambda^{(1)}_{S_2}\lambda^{(A)}_{S_2}$	8
${\lambda^{(A)}_{S_2}}^2$	16
$\lambda_{X_1}(\lambda_{X_1})^*$	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0
$\lambda^{(1)}_{X_7}(\lambda^{(1)}_{X_7})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0
$g_1^2(\frac{27g_2^2}{2} - 27\lambda^{(1)}_{S_2})$	0
${\lambda^{(A)}_{12}}^2$	0
${\lambda^{(A)}_{HS_2}}^2$	0
$\lambda_8(\lambda_8)^*$	0
$\lambda_9(\lambda_9)^*$	0
$\lambda_{10}(\lambda_{10})^*$	0
$g_1^2(\frac{3g_2^2}{2} - 3\lambda^{(1)}_{S_2})$	0
$\lambda_{X_{10}}(\lambda_{X_{10}})^*$	0
$g_1^2\lambda^{(1)}_{S_2}$	0
$\lambda^{(1)}_{X_5}(3(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0
$\lambda^{(2)}_{X_5}(8(\lambda^{(1)}_{X_5})^* + \frac{20(\lambda^{(2)}_{X_5})^*}{3})$	0
$\lambda^{(1)}_{12}\lambda^{(\times)}_{12}$	8
${\lambda^{(\times)}_{12}}^2$	$\frac{20}{3}$
$g_1^2(32g_2^2 - 48\lambda^{(1)}_{S_2})$	0

$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(\frac{8(\lambda_{X_7}^{(1)})^*}{3} + 4(\lambda_{X_7}^{(2)})^*)$	0

$\beta^{(1)}_{\lambda_{S_2}^{(A)}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
g_2^4	-2	-2	-2
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	1	0	1
$g_2^2\lambda_{S_2}^{(A)}$	-24	-24	-24
${\lambda_{HS_2}^{(A)}}^2$	3	3	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12	12	12
${\lambda_{S_2}^{(A)}}^2$	20	20	20
$\lambda_{X_1}(\lambda_{X_1})^*$	0	$-\frac{224}{3}$	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	$-\frac{44}{3}$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	$-\frac{4}{3}$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	$-\frac{8}{3}$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{S_2}^{(A)}}^2$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
${\lambda_{12}^{(A)}}^2$	0	0	0
${\lambda_{12}^{(\times)}}^2$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2\lambda_{S_2}^{(A)}$	0	0	0
${\lambda_{HS_2}^{(A)}}^2$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0

$\lambda_{S_2}^{(A)^2}$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{12}^{(A)^2}$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
g_2^4	0	0	0
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	0	0	0
$g_2^2\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{HS_2}^{(A)^2}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	0	0	0
$\lambda_{S_2}^{(A)^2}$	0	0	0
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0	0	0
$\lambda_{12}^{(A)^2}$	0	0	0
$\lambda_{12}^{(\times)^2}$	0	0	0
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
g_2^4	-2	-2	-2
$g_1^2(8g_2^2 - 12\lambda_{S_2}^{(A)})$	1	0	0
$g_2^2\lambda_{S_2}^{(A)}$	-24	-24	-24
$\lambda_{HS_2}^{(A)^2}$	3	3	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12	12	12
$\lambda_{S_2}^{(A)^2}$	20	20	20
$\lambda_{X_1}(\lambda_{X_1})^*$	0	0	$-\frac{224}{3}$
$\lambda_{X_2}(\lambda_{X_2})^*$	0	0	$-\frac{44}{3}$
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0	0	$-\frac{8}{3}$
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	$-\frac{4}{3}$	0	0
$\lambda_{12}^{(A)^2}$	12	12	12
$\lambda_{12}^{(\times)^2}$	$-\frac{4}{3}$	$-\frac{4}{3}$	$-\frac{4}{3}$
$g_1^2(32g_2^2 - 48\lambda_{S_2}^{(A)})$	0	1	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0	0	$-\frac{4}{3}$
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	$-\frac{4}{3}$

$\mathcal{S}$	E ($\alpha=-4$)
g_2^4	-2
$g_1^2(8g_2^2-12\lambda_{S_2}^{(A)})$	1
$g_2^2\lambda_{S_2}^{(A)}$	-24
${\lambda_{HS_2}^{(A)}}^2$	3
$\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)}$	12
${\lambda_{S_2}^{(A)}}^2$	20
$\lambda_{X_1}(\lambda_{X_1})^*$	0
$\lambda_{X_2}(\lambda_{X_2})^*$	0
$\lambda_{X_7}^{(1)}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{16}})^*$	0
$\lambda_{X_5}^{(2)}(\lambda_{X_5}^{(2)})^*$	0
${\lambda_{12}^{(A)}}^2$	12
${\lambda_{12}^{(\times)}}^2$	$-\frac{4}{3}$
$g_1^2(32g_2^2-48\lambda_{S_2}^{(A)})$	0
$\lambda_{X_7}^{(2)}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_7}^{(1)}(3(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0

$\beta_{\lambda_5}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	1	1	1
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{33}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	4	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	8	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0	0	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_5$	-9	-9	-9
$g_1^2\lambda_5$	-9	-3	-9
$\lambda_{12}^{(1)}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4	4	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4	4	4
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	1	1	1

$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda^{(A)}_{HS_1})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda^{(1)}_{X_5})^*-(\lambda^{(2)}_{X_5})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{HS_2})$	0	0	0
$(\lambda_{X_{14}})^*((\lambda^{(1)}_{X_5})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_5$	-9	-9	-9
$g_1^2\lambda_5$	-9	-3	-9
$\lambda^{(1)}_{12}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda^{(1)}_{HS_1}\lambda_5$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda_5$	4	4	4
$\lambda^{(1)}_{X_7}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda^{(A)}_{12}}{6}+\frac{2\lambda^{(A)}_{HS_1}}{3}+2\lambda^{(A)}_{HS_2})$	1	1	1
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda^{(A)}_{HS_1})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda^{(1)}_{X_5})^*-(\lambda^{(2)}_{X_5})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{HS_2})$	0	0	0
$(\lambda_{X_{14}})^*((\lambda^{(1)}_{X_5})^*+2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_5$	$-\frac{21}{2}$	$-\frac{21}{2}$	$-\frac{21}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda^{(1)}_{12}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda^{(1)}_{HS_1}\lambda_5$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda_5$	4	4	4
$\lambda^{(1)}_{X_7}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	8
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda^{(A)}_{12}}{6}+\frac{2\lambda^{(A)}_{HS_1}}{3}+2\lambda^{(A)}_{HS_2})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda^{(A)}_{HS_1})$	1	1	1
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda^{(A)}_{12}}{3}-2\lambda^{(\times)}_{12}-2\lambda^{(A)}_{HS_1}-\frac{4\lambda^{(A)}_{HS_2}}{3})$	0	0	0
$(\lambda_{X_{14}})^*(2(\lambda^{(1)}_{X_5})^*-(\lambda^{(2)}_{X_5})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda^{(2)}_{X_7}(\lambda_{X_{12}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda^{(A)}_{HS_2})$	0	0	0
$(\lambda_{X_{14}})^*((\lambda^{(1)}_{X_5})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_5$	$-\frac{33}{2}$	$-\frac{33}{2}$	$-\frac{33}{2}$
$g_1^2\lambda_5$	$-\frac{9}{2}$	$-\frac{33}{2}$	$-\frac{9}{2}$
$\lambda^{(1)}_{12}\lambda_5$	2	2	2
$\lambda_1\lambda_5$	2	2	2
$\lambda^{(1)}_{HS_1}\lambda_5$	4	4	4
$\lambda^{(1)}_{HS_2}\lambda_5$	4	4	4
$\lambda^{(1)}_{X_7}(\lambda_{X_{12}})^*$	0	0	4
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0	0	8

$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0	0	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	1	1	1
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	2	0	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0	0	−2

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+4\lambda_{HS_2}^{(A)})$	0
$(\lambda_{X_{14}})^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	0
$g_2^2\lambda_5$	$-\frac{33}{2}$
$g_1^2\lambda_5$	$-\frac{33}{2}$
$\lambda_{12}^{(1)}\lambda_5$	2
$\lambda_1\lambda_5$	2
$\lambda_{HS_1}^{(1)}\lambda_5$	4
$\lambda_{HS_2}^{(1)}\lambda_5$	4
$\lambda_{X_7}^{(1)}(\lambda_{X_{12}})^*$	0
$\lambda_{X_{16}}(\lambda_{X_{12}})^*$	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)})$	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-4\lambda_{HS_1}^{(A)})$	0
$\lambda_5(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger)-\frac{4\lambda_{12}^{(A)}}{3}-2\lambda_{12}^{(\times)}-2\lambda_{HS_1}^{(A)}-\frac{4\lambda_{HS_2}^{(A)}}{3})$	1
$(\lambda_{X_{14}})^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0
$\lambda_{X_7}^{(2)}(\lambda_{X_{12}})^*$	0

$\beta_{\lambda_{X_1}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$g_2^2\lambda_{X_1}$	0	−24	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	12	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	−8	0
${\lambda_{X_2}}^2$	0	11	0
${\lambda_{X_7}^{(1)}}^2$	0	1	0
${\lambda_{X_{16}}}^2$	0	2	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	0
${\lambda_{X_7}^{(1)}}^2$	0	0	0
${\lambda_{X_{16}}}^2$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	0

$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	0
${\lambda_{X_7}^{(1)}}^2$	0	0	0
${\lambda_{X_{16}}}^2$	0	0	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$g_2^2\lambda_{X_1}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	12
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	0
${\lambda_{X_2}}^2$	0	0	9
${\lambda_{X_7}^{(1)}}^2$	0	0	3
${\lambda_{X_{16}}}^2$	0	0	2
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
${\lambda_{X_7}^{(2)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$g_2^2\lambda_{X_1}$	0	0	-24
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0	0	12
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0	0	-8
${\lambda_{X_2}}^2$	0	0	11
${\lambda_{X_7}^{(1)}}^2$	0	0	3
${\lambda_{X_{16}}}^2$	0	0	2
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
${\lambda_{X_7}^{(2)}}^2$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$g_2^2\lambda_{X_1}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_1}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_1}$	0
${\lambda_{X_2}}^2$	0
${\lambda_{X_7}^{(1)}}^2$	0
${\lambda_{X_{16}}}^2$	0
$\lambda_{X_7}^{(1)}\lambda_{X_7}^{(2)}$	0
${\lambda_{X_7}^{(2)}}^2$	0

$\beta_{\lambda_{X_2}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0	88	0
$g_2^2\lambda_{X_2}$	0	-24	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	2	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	4	0
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0	22	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0	0	0
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0	0	0
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0	0	72
$g_2^2\lambda_{X_2}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	6
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0	0	18
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0	0	88
$g_2^2\lambda_{X_2}$	0	0	-24
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0	0	22
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_1}(\lambda_{X_2})^*$	0
$g_2^2\lambda_{X_2}$	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(1)}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{16}}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_2}$	0
$\lambda_{12}^{(1)}(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_7}^{(1)}+4\lambda_{X_7}^{(2)})$	0

$\beta_{\lambda_{X_5}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda^{(1)}_{X_5})^*$	24	0	0
$g_2^2\lambda^{(1)}_{X_5}$	-12	0	0
$g_1^2\lambda^{(1)}_{X_5}$	-6	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 6\lambda_{X_9})$	1	0	0
$\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$	4	0	0
$\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	2	0	0
$\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	8	0	0
$\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$	4	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 10\lambda_{X_9})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda_{X_6}(5(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5} + 4\lambda_{X_9})$	0	0	0
$\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$	0	0	0
$\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0
$\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda^{(1)}_{X_5})^*$	0	0	0
$g_2^2\lambda^{(1)}_{X_5}$	0	0	0
$g_1^2\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 6\lambda_{X_9})$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 10\lambda_{X_9})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda_{X_6}(5(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0
$\lambda^{(\times)}_{12}(4\lambda^{(2)}_{X_5} + 4\lambda_{X_9})$	0	0	0
$\lambda^{(A)}_{S_1}(-4\lambda^{(1)}_{X_5} - 2\lambda^{(2)}_{X_5})$	0	0	0
$\lambda^{(1)}_{S_2}(8\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0
$\lambda^{(A)}_{S_2}(4\lambda^{(1)}_{X_5} + 2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda^{(1)}_{X_5})^*$	0	0	0
$g_2^2\lambda^{(1)}_{X_5}$	0	0	0
$g_1^2\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 6\lambda_{X_9})$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{12}(8\lambda^{(1)}_{X_5} + 10\lambda_{X_9})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda_{X_6}(5(\lambda^{(1)}_{X_5})^* + (\lambda^{(2)}_{X_5})^*)$	0	0	0

$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	40	0	0
$g_2^2\lambda_{X_5}^{(1)}$	-12	0	0
$g_1^2\lambda_{X_5}^{(1)}$	-6	0	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	4	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	-4	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	2	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(1)}$	-24	0	0
$g_1^2\lambda_{X_5}^{(1)}$	-6	0	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	8	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	1	0	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda_{X_5}^{(1)})^*$	0
$g_2^2\lambda_{X_5}^{(1)}$	0
$g_1^2\lambda_{X_5}^{(1)}$	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+6\lambda_{X_9})$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(1)}$	0

$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(1)}$	0
$\lambda_{12}^{(1)}(8\lambda_{X_5}^{(1)}+10\lambda_{X_9})$	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(1)}$	0
$\lambda_{X_6}(5(\lambda_{X_5}^{(1)})^*+(\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(2)}+4\lambda_{X_9})$	0
$\lambda_{S_1}^{(A)}(-4\lambda_{X_5}^{(1)}-2\lambda_{X_5}^{(2)})$	0
$\lambda_{S_2}^{(1)}(8\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0
$\lambda_{S_2}^{(A)}(4\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0

$\beta_{\lambda_{X_5}^{(2)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(\times)}(16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(\times)}(16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_5}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_5}^{(2)}$	0	0	0
$g_1^2\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{12}^{(\times)}(16\lambda_{X_5}^{(1)}+20\lambda_{X_5}^{(2)}+8\lambda_{X_9})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_5}^{(2)}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda^{(2)}_{X_5})^*$	0	0	0
$g_2^2\lambda^{(2)}_{X_5}$	0	0	0
$g_1^2\lambda^{(2)}_{X_5}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_5}$	0	0	0
$\lambda^{(\times)}_{12}(16\lambda^{(1)}_{X_5} + 20\lambda^{(2)}_{X_5} + 8\lambda_{X_9})$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	-2	0	0
$\lambda_{X_6}(\lambda^{(2)}_{X_5})^*$	16	0	0
$g_2^2\lambda^{(2)}_{X_5}$	-24	0	0
$g_1^2\lambda^{(2)}_{X_5}$	-6	0	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_5}$	8	0	0
$\lambda^{(\times)}_{12}(16\lambda^{(1)}_{X_5} + 20\lambda^{(2)}_{X_5} + 8\lambda_{X_9})$	1	0	0
$\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$	2	0	0
$\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$	2	0	0
$\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$	2	0	0
$\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$	-2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda^{(2)}_{X_5})^*$	0
$g_2^2\lambda^{(2)}_{X_5}$	0
$g_1^2\lambda^{(2)}_{X_5}$	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_5}$	0
$\lambda^{(\times)}_{12}(16\lambda^{(1)}_{X_5} + 20\lambda^{(2)}_{X_5} + 8\lambda_{X_9})$	0
$\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_5}$	0
$\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_5}$	0
$\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_5}$	0
$\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_5}$	0

$\beta^{(1)}_{\lambda_{X_6}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda^{(1)}_{S_1}\lambda_{X_6}$	12	0	0
${\lambda^{(1)}_{X_5}}^2$	3	0	0
${\lambda_{X_9}}^2$	9	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_6}$	0	0	0
$\lambda^{(1)}_{X_5}\lambda^{(2)}_{X_5}$	0	0	0
${\lambda^{(2)}_{X_5}}^2$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0	0	0
${\lambda_{X_5}^{(1)}}^2$	0	0	0
${\lambda_{X_9}}^2$	0	0	0
${\lambda_{X_{18}}}^2$	0	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0	0	0
${\lambda_{X_5}^{(1)}}^2$	0	0	0
${\lambda_{X_9}}^2$	0	0	0
${\lambda_{X_{18}}}^2$	0	0	0
$g_2^2\lambda_{X_6}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	12	0	0
${\lambda_{X_5}^{(1)}}^2$	1	0	0
${\lambda_{X_9}}^2$	11	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$g_2^2\lambda_{X_6}$	-24	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	-8	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0	0	0
${\lambda_{X_5}^{(2)}}^2$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	12	0	0
${\lambda_{X_5}^{(1)}}^2$	3	0	0
${\lambda_{X_9}}^2$	11	0	0
${\lambda_{X_{18}}}^2$	2	0	0
$g_2^2\lambda_{X_6}$	-24	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_6}$	-8	0	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	2	0	0
${\lambda_{X_5}^{(2)}}^2$	1	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{S_1}^{(1)}\lambda_{X_6}$	0
${\lambda_{X_5}^{(1)}}^2$	0
${\lambda_{X_9}}^2$	0
${\lambda_{X_{18}}}^2$	0
$g_2^2\lambda_{X_6}$	0

$\lambda_{S_1}^{(A)}\lambda_{X_6}$	0
$\lambda_{X_5}^{(1)}\lambda_{X_5}^{(2)}$	0
${\lambda_{X_5}^{(2)}}^2$	0

$\beta_{\lambda_{X_7}^{(1)}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	40	0
$g_2^2\lambda_{X_7}^{(1)}$	0	-12	0
$g_1^2\lambda_{X_7}^{(1)}$	0	-6	0
$\lambda_{12}^{(1)}(10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	1	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	4	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	4	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	2	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	-4	0
$\lambda_5\lambda_{X_{12}}$	0	4	0
$\lambda_{12}^{(1)}(6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	0
$g_1^2\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	0
$g_1^2\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{12}^{(1)}(10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	0

$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	24
$g_2^2\lambda_{X_7}^{(1)}$	0	0	-12
$g_1^2\lambda_{X_7}^{(1)}$	0	0	-6
$\lambda_{12}^{(1)}(10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	8
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	4
$\lambda_{12}^{(1)}(6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	1
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	4
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_7}^{(1)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(1)}$	0	0	-24
$g_1^2\lambda_{X_7}^{(1)}$	0	0	-6
$\lambda_{12}^{(1)}(10\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	1
$\lambda_{HS_1}^{(1)}\lambda_{X_{16}}$	0	0	4
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(1)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	2
$\lambda_{12}^{(1)}(6\lambda_{X_2}+8\lambda_{X_7}^{(1)})$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(1)}$	0	0	0
$\lambda_{X_1}(5(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	8
$\lambda_{12}^{(\times)}(4\lambda_{X_2}+4\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_1}^{(1)}(8\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_1}^{(A)}(4\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_2}^{(A)}(-4\lambda_{X_7}^{(1)}-2\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)

$\lambda_{X_1}(\lambda^{(1)}_{X_7})^*$	0
$g_2^2\lambda^{(1)}_{X_7}$	0
$g_1^2\lambda^{(1)}_{X_7}$	0
$\lambda^{(1)}_{12}(10\lambda_{X_2}+8\lambda^{(1)}_{X_7})$	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{16}}$	0
$\lambda^{(1)}_{S_1}\lambda^{(1)}_{X_7}$	0
$\lambda^{(1)}_{S_2}\lambda^{(1)}_{X_7}$	0
$\lambda^{(A)}_{S_2}\lambda^{(1)}_{X_7}$	0
$\lambda_5\lambda_{X_{12}}$	0
$\lambda^{(1)}_{12}(6\lambda_{X_2}+8\lambda^{(1)}_{X_7})$	0
$\lambda^{(A)}_{S_1}\lambda^{(1)}_{X_7}$	0
$\lambda_{X_1}(5(\lambda^{(1)}_{X_7})^*+(\lambda^{(2)}_{X_7})^*)$	0
$\lambda^{(\times)}_{12}(4\lambda_{X_2}+4\lambda^{(2)}_{X_7})$	0
$\lambda^{(1)}_{S_1}(8\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0
$\lambda^{(A)}_{S_1}(4\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0
$\lambda^{(A)}_{S_2}(-4\lambda^{(1)}_{X_7}-2\lambda^{(2)}_{X_7})$	0

$\beta^{(1)}_{\lambda^{(2)}_{X_7}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_1}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_2^2\lambda^{(2)}_{X_7}$	0	0	0
$g_1^2\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(\times)}_{12}(8\lambda_{X_2}+16\lambda^{(1)}_{X_7}+20\lambda^{(2)}_{X_7})$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_1}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_2^2\lambda^{(2)}_{X_7}$	0	0	0
$g_1^2\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(\times)}_{12}(8\lambda_{X_2}+16\lambda^{(1)}_{X_7}+20\lambda^{(2)}_{X_7})$	0	0	0
$\lambda^{(1)}_{S_1}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(A)}_{S_1}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(1)}_{S_2}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_1}(\lambda^{(2)}_{X_7})^*$	0	0	0
$g_2^2\lambda^{(2)}_{X_7}$	0	0	0
$g_1^2\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(2)}_{X_7}$	0	0	0
$\lambda^{(\times)}_{12}(8\lambda_{X_2}+16\lambda^{(1)}_{X_7}+20\lambda^{(2)}_{X_7})$	0	0	0

$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	0
$g_2^2\lambda_{X_7}^{(2)}$	0	0	0
$g_1^2\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{12}^{(\times)}(8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0	0	16
$g_2^2\lambda_{X_7}^{(2)}$	0	0	-24
$g_1^2\lambda_{X_7}^{(2)}$	0	0	-6
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0	0	8
$\lambda_{12}^{(\times)}(8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0	0	1
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0	0	-2
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0	0	2
$\lambda_5\lambda_{X_{12}}$	0	0	-2

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_1}(\lambda_{X_7}^{(2)})^*$	0
$g_2^2\lambda_{X_7}^{(2)}$	0
$g_1^2\lambda_{X_7}^{(2)}$	0
$\lambda_{12}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{12}^{(\times)}(8\lambda_{X_2}+16\lambda_{X_7}^{(1)}+20\lambda_{X_7}^{(2)})$	0
$\lambda_{S_1}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_1}^{(A)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_7}^{(2)}$	0
$\lambda_{S_2}^{(A)}\lambda_{X_7}^{(2)}$	0
$\lambda_5\lambda_{X_{12}}$	0

$\beta_{\lambda_{X_9}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_6}(\lambda_{X_9})^*$	72	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	6	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0

$\lambda_{S_1}^{(1)}\lambda_{X_9}$	18	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_6}(\lambda_{X_9})^*$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_9}$	0	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_6}(\lambda_{X_9})^*$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_9}$	0	0	0
$g_2^2\lambda_{X_9}$	0	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_6}(\lambda_{X_9})^*$	88	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_9}$	22	0	0
$g_2^2\lambda_{X_9}$	-24	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_6}(\lambda_{X_9})^*$	88	0	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_9}$	22	0	0
$g_2^2\lambda_{X_9}$	-24	0	0
$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)}+4\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_6}(\lambda_{X_9})^*$	0
$\lambda_{12}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_9}$	0
$g_2^2\lambda_{X_9}$	0

$\lambda_{12}^{(1)}(6\lambda_{X_5}^{(1)} + 2\lambda_{X_5}^{(2)})$	0
$\lambda_{12}^{(\times)}(4\lambda_{X_5}^{(1)} + 4\lambda_{X_5}^{(2)})$	0

$\beta_{\lambda_{X_{10}}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$g_2^2\lambda_{X_{10}}$	0	0	-9
$g_1^2\lambda_{X_{10}}$	0	0	-3
$\lambda_1\lambda_{X_{10}}$	0	0	2
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	8
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	-4
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	2

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$g_2^2\lambda_{X_{10}}$	0	0	-9
$g_1^2\lambda_{X_{10}}$	0	0	-3
$\lambda_1\lambda_{X_{10}}$	0	0	2
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	8
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	-4
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	2

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{10}}$	0	0	0
$g_1^2\lambda_{X_{10}}$	0	0	0
$\lambda_1\lambda_{X_{10}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0	0	0

$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0	0	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_{10}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_2^2\lambda_{X_{10}}$	0
$g_1^2\lambda_{X_{10}}$	0
$\lambda_1\lambda_{X_{10}}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{10}}$	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{10}}$	0
$\lambda_{S_2}^{(1)}\lambda_{X_{10}}$	0

$\beta_{\lambda_{X_{12}}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	4	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	8	0
$g_2^2\lambda_{X_{12}}$	0	$-\frac{21}{2}$	0
$g_1^2\lambda_{X_{12}}$	0	$-\frac{9}{2}$	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	2	0
$\lambda_1\lambda_{X_{12}}$	0	2	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	4	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	4	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	-4	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	0
$g_2^2\lambda_{X_{12}}$	0	0	0
$g_1^2\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_1\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0

$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	0
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	0
$g_2^2\lambda_{X_{12}}$	0	0	0
$g_1^2\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_1\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	4
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	8
$g_2^2\lambda_{X_{12}}$	0	0	$-\frac{21}{2}$
$g_1^2\lambda_{X_{12}}$	0	0	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	2
$\lambda_1\lambda_{X_{12}}$	0	0	2
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	-4
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0	0	4
$\lambda_{X_{16}}(\lambda_5)^*$	0	0	8
$g_2^2\lambda_{X_{12}}$	0	0	$-\frac{33}{2}$
$g_1^2\lambda_{X_{12}}$	0	0	$-\frac{9}{2}$
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0	0	2
$\lambda_1\lambda_{X_{12}}$	0	0	2
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0	0	4
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0	0	-2
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0	0	-2
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0	0	-2

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_{12}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_7}^{(1)}(\lambda_5)^*$	0
$\lambda_{X_{16}}(\lambda_5)^*$	0

$g_2^2\lambda_{X_{12}}$	0
$g_1^2\lambda_{X_{12}}$	0
$\lambda_{12}^{(1)}\lambda_{X_{12}}$	0
$\lambda_1\lambda_{X_{12}}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{12}}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{12}}$	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{12}}$	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{12}}$	0
$\lambda_{X_7}^{(2)}(\lambda_5)^*$	0
$\lambda_{12}^{(A)}\lambda_{X_{12}}$	0
$\lambda_{12}^{(\times)}\lambda_{X_{12}}$	0

$\beta_{\lambda_{X_{14}}}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_{X_{14}}$	$-\frac{21}{2}$	0	0
$g_1^2\lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	2	0	0
$\lambda_1\lambda_{X_{14}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_{X_{14}}$	0	0	0
$g_1^2\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_1\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^*-(\lambda_{X_5}^{(2)})^*+4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^*+2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_{X_{14}}$	0	0	0
$g_1^2\lambda_{X_{14}}$	0	0	0

$\lambda_{12}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_1\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	4	0	0
$g_2^2\lambda_{X_{14}}$	$-\frac{21}{2}$	0	0
$g_1^2\lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	2	0	0
$\lambda_1\lambda_{X_{14}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	4	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0	0	0
$g_2^2\lambda_{X_{14}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{14}}$	$-\frac{9}{2}$	0	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	2	0	0
$\lambda_1\lambda_{X_{14}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	4	0	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	$-\frac{4}{3}$	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	$-\frac{4}{3}$	0	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	2	0	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	-2	0	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	-2	0	0

$\mathcal{S}$	E ($\alpha = -4$)
$\lambda_{X_{14}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*((\lambda_{X_5}^{(1)})^* + 2(\lambda_{X_{18}})^*)$	0
$g_2^2\lambda_{X_{14}}$	0
$g_1^2\lambda_{X_{14}}$	0
$\lambda_{12}^{(1)}\lambda_{X_{14}}$	0
$\lambda_1\lambda_{X_{14}}$	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{14}}$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_{14}}$	0
$\lambda_{HS_2}^{(A)}\lambda_{X_{14}}$	0

$\lambda_{HS_1}^{(A)}\lambda_{X_{14}}$	0
$(\lambda_5)^*(2(\lambda_{X_5}^{(1)})^* - (\lambda_{X_5}^{(2)})^* + 4(\lambda_{X_{18}})^*)$	0
$\lambda_{12}^{(A)}\lambda_{X_{14}}$	0
$\lambda_{12}^{(\times)}\lambda_{X_{14}}$	0

$\beta_{\lambda_{X_{15}}}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0
$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$g_2^2\lambda_{X_{15}}$	-9	0	0
$g_1^2\lambda_{X_{15}}$	-3	0	0
$\lambda_1\lambda_{X_{15}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	8	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	2	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$g_2^2\lambda_{X_{15}}$	-9	0	0
$g_1^2\lambda_{X_{15}}$	-3	0	0
$\lambda_1\lambda_{X_{15}}$	2	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	8	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	4	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	2	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0
$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(1)}\lambda_{X_{15}}$	0	0	0
$\lambda_{HS_1}^{(A)}\lambda_{X_{15}}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{15}}$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger) + 2\,\mathrm{Tr}(Y_e(Y_e)^\dagger) + 6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$g_2^2\lambda_{X_{15}}$	0	0	0
$g_1^2\lambda_{X_{15}}$	0	0	0

$\lambda_1\lambda_{X_{15}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{15}}$	0	0	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{15}}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{15}}$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_{15}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$g_2^2\lambda_{X_{15}}$	0
$g_1^2\lambda_{X_{15}}$	0
$\lambda_1\lambda_{X_{15}}$	0
$\lambda^{(1)}_{HS_1}\lambda_{X_{15}}$	0
$\lambda^{(A)}_{HS_1}\lambda_{X_{15}}$	0
$\lambda^{(1)}_{S_1}\lambda_{X_{15}}$	0

$\beta^{(1)}_{\lambda_{X_{16}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	40	0
$g_2^2\lambda_{X_{16}}$	0	$-\frac{33}{2}$	0
$g_1^2\lambda_{X_{16}}$	0	$-\frac{3}{2}$	0
$\lambda_1\lambda_{X_{16}}$	0	6	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	2	0
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0	1	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	2	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	-4	0
$\lambda_5\lambda_{X_{12}}$	0	4	0
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	0
$g_2^2\lambda_{X_{16}}$	0	0	0
$g_1^2\lambda_{X_{16}}$	0	0	0
$\lambda_1\lambda_{X_{16}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	0
$g_2^2\lambda_{X_{16}}$	0	0	0

$g_1^2\lambda_{X_{16}}$	0	0	0
$\lambda_1\lambda_{X_{16}}$	0	0	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	24
$g_2^2\lambda_{X_{16}}$	0	0	$-\frac{9}{2}$
$g_1^2\lambda_{X_{16}}$	0	0	$-\frac{3}{2}$
$\lambda_1\lambda_{X_{16}}$	0	0	6
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	6
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	2
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	0
$\lambda_5\lambda_{X_{12}}$	0	0	12
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	1
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	1
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0	0	40
$g_2^2\lambda_{X_{16}}$	0	0	$-\frac{33}{2}$
$g_1^2\lambda_{X_{16}}$	0	0	$-\frac{3}{2}$
$\lambda_1\lambda_{X_{16}}$	0	0	6
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0	0	0
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	1
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0	0	2
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0	0	-4
$\lambda_5\lambda_{X_{12}}$	0	0	4
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_{16}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_{X_1}(\lambda_{X_{16}})^*$	0
$g_2^2\lambda_{X_{16}}$	0
$g_1^2\lambda_{X_{16}}$	0
$\lambda_1\lambda_{X_{16}}$	0
$\lambda^{(1)}_{HS_1}\lambda^{(1)}_{X_7}$	0
$\lambda^{(1)}_{HS_2}(10\lambda_{X_2}+8\lambda_{X_{16}})$	0
$\lambda^{(1)}_{S_2}\lambda_{X_{16}}$	0
$\lambda^{(A)}_{S_2}\lambda_{X_{16}}$	0
$\lambda_5\lambda_{X_{12}}$	0
$\lambda^{(1)}_{HS_2}(6\lambda_{X_2}+8\lambda_{X_{16}})$	0

$\lambda^{(1)}_{HS_1}(6\lambda^{(1)}_{X_7}+2\lambda^{(2)}_{X_7})$	0

$\beta^{(1)}_{\lambda^{(1)}_{X_{17}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	0	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	0	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0	0	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	1	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	-9	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	-3	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	2	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0	1	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	1	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0	1	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0	1	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0	1	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0	1	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	1	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	1	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	-9	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	-3	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	2	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0	1	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	1	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0	1	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0	1	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0	1	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0	1	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	0	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	0	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0	0	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0	0	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0	0	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0	0	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0	0	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0	0	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0	0	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0	0	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0	0	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda^{(1)}_{X_{17}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$\lambda_9(8(\lambda^{(1)}_{X_{17}})^*-4(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})^*)$	0
$g_2^2\lambda^{(1)}_{X_{17}}$	0
$g_1^2\lambda^{(1)}_{X_{17}}$	0
$\lambda^{(1)}_{12}\lambda^{(1)}_{X_{17}}$	0
$\lambda^{(A)}_{12}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda^{(1)}_{X_{17}}}{2})$	0
$\lambda_1(4\lambda^{(1)}_{X_{17}}-2\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger})$	0
$\lambda^{(1)}_{HS_1}(4\lambda^{(1)}_{X_{17}}-6\lambda_{11})$	0
$\lambda^{(A)}_{HS_1}(-\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}+\frac{\lambda_{11}}{2}-2\lambda^{(1)}_{X_{17}})$	0
$\lambda^{(1)}_{HS_2}(4\lambda^{(1)}_{X_{17}}-6\lambda_{10})$	0
$\lambda^{(A)}_{HS_2}(\lambda^{\mathrm{Cross}}_{HH^\dagger S_1^\dagger S_2^\dagger}-\frac{\lambda_{10}}{2}+2\lambda^{(1)}_{X_{17}})$	0

$\beta^{(1)}_{\lambda_{X_{18}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	12	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	24	0	0

$g_2^2\lambda_{X_{18}}$	$-\frac{9}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0
$\lambda^{(1)}_{HS_1}(6\lambda_{X_9}+8\lambda_{X_{18}})$	1	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	6	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	2	0	0
$\lambda^{(1)}_{HS_1}(10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0	0	0
$g_2^2\lambda_{X_{18}}$	0	0	0
$g_1^2\lambda_{X_{18}}$	0	0	0
$\lambda_1\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0	0	0
$g_2^2\lambda_{X_{18}}$	0	0	0
$g_1^2\lambda_{X_{18}}$	0	0	0
$\lambda_1\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	0	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_1}(10\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	0	0	0
$\lambda^{(1)}_{HS_2}(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	40	0	0
$g_2^2\lambda_{X_{18}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0
$\lambda^{(1)}_{HS_1}(6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda^{(1)}_{HS_2}\lambda^{(1)}_{X_5}$	2	0	0
$\lambda^{(1)}_{S_1}\lambda_{X_{18}}$	2	0	0
$\lambda^{(1)}_{HS_1}(10\lambda_{X_9}+8\lambda_{X_{18}})$	1	0	0
$\lambda^{(A)}_{S_1}\lambda_{X_{18}}$	-4	0	0

$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	1	0	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	4	0	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	40	0	0
$g_2^2\lambda_{X_{18}}$	$-\frac{33}{2}$	0	0
$g_1^2\lambda_{X_{18}}$	$-\frac{3}{2}$	0	0
$\lambda_1\lambda_{X_{18}}$	6	0	0
$\lambda_{HS_1}^{(1)}(6\lambda_{X_9}+8\lambda_{X_{18}})$	0	0	0
$\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$	0	0	0
$\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	2	0	0
$\lambda_{HS_1}^{(1)}(10\lambda_{X_9}+8\lambda_{X_{18}})$	1	0	0
$\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	-4	0	0
$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_{18}}(6\,\mathrm{Tr}(Y_d(Y_d)^\dagger)+2\,\mathrm{Tr}(Y_e(Y_e)^\dagger)+6\,\mathrm{Tr}(Y_u(Y_u)^\dagger))$	0
$(\lambda_5)^*(\lambda_{X_{14}})^*$	0
$\lambda_{X_6}(\lambda_{X_{18}})^*$	0
$g_2^2\lambda_{X_{18}}$	0
$g_1^2\lambda_{X_{18}}$	0
$\lambda_1\lambda_{X_{18}}$	0
$\lambda_{HS_1}^{(1)}(6\lambda_{X_9}+8\lambda_{X_{18}})$	0
$\lambda_{HS_2}^{(1)}\lambda_{X_5}^{(1)}$	0
$\lambda_{S_1}^{(1)}\lambda_{X_{18}}$	0
$\lambda_{HS_1}^{(1)}(10\lambda_{X_9}+8\lambda_{X_{18}})$	0
$\lambda_{S_1}^{(A)}\lambda_{X_{18}}$	0
$\lambda_{HS_2}^{(1)}(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0

$\beta_{m_1^2}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda_{X_9})^*$	12	0	0
$\lambda_{X_9}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	12	0	0
$\lambda_{12}^{(1)}m_2^2$	6	6	6
$\lambda_{S_1}^{(1)}m_1^2$	4	4	4
$\mu_{\mathrm{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	12	0
$\lambda_{X_7}^{(1)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	12	0
$g_1^2m_1^2$	0	-6	-24
$g_2^2m_1^2$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(8\lambda_{S_1}^{(1)}+4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\mathrm{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^*+(\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda_{12}^{(1)} m_2^2$	4	4	4
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	6	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_9})^*$	0	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda_{12}^{(1)} m_2^2$	4	4	4
$\lambda_{S_1}^{(1)} m_1^2$	6	6	6
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_1^2 m_1^2$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{27}{2}$
$g_2^2 m_1^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	6	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	0	0	0
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_9})^*$	20	0	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	20	0	0
$\lambda_{12}^{(1)} m_2^2$	2	2	2
$\lambda_{S_1}^{(1)} m_1^2$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	4
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	4
$g_1^2 m_1^2$	0	-6	-6
$g_2^2 m_1^2$	-12	-12	-12
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1	1	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_9})^*$	20	0	0

$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	20	0	0
$\lambda_{12}^{(1)} m_2^2$	0	0	0
$\lambda_{S_1}^{(1)} m_1^2$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0	0	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	12
$g_1^2 m_1^2$	0	-6	-6
$g_2^2 m_1^2$	-12	-12	-12
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0	0	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1	1	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0	0	4
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	4

$\mathcal{S}$	E ($\alpha = -4$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_9})^*$	0
$\lambda_{X_9}(\mu_{\text{T3GeneralS1Sq}})^*$	0
$\lambda_{12}^{(1)} m_2^2$	0
$\lambda_{S_1}^{(1)} m_1^2$	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_7}^{(1)})^*$	0
$\lambda_{X_7}^{(1)}(\mu_{\text{T3GeneralS2Sq}})^*$	0
$g_1^2 m_1^2$	-24
$g_2^2 m_1^2$	-12
$\mu_{\text{T3GeneralS1S2}}(\lambda_{11})^*$	0
$\lambda_{11}(\mu_{\text{T3GeneralS1S2}})^*$	0
$m_1^2(8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	1
$m_2^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1
$\mu_{\text{T3GeneralS2Sq}}(3(\lambda_{X_7}^{(1)})^* + (\lambda_{X_7}^{(2)})^*)$	0
$\lambda_{X_7}^{(2)}(\mu_{\text{T3GeneralS2Sq}})^*$	0

$\beta_{m_2^2}^{(1)}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	4	0	0
$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	4	0	0
$g_2^2 m_2^2$	-12	-12	-12
$g_1^2 m_2^2$	-6	0	-6
$\lambda_{12}^{(1)} m_1^2$	2	2	2
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	1	1	1
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0	20	0
$\lambda_{X_2}(\mu_{\text{T3GeneralS2Sq}})^*$	0	20	0
$\lambda_{S_2}^{(1)} m_2^2$	0	0	0
$\mu_{\text{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0	0	0

$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	4	4	4
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_2}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6
$\mu_{\text{T3GeneralS1S2}}(\lambda_{10})^*$	0	6	0
$\lambda_{10}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$\mu_{\text{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$g_2^2 m_2^2$	$-\frac{9}{2}$	$-\frac{9}{2}$	$-\frac{9}{2}$
$g_1^2 m_2^2$	$-\frac{27}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$\lambda_{12}^{(1)} m_1^2$	4	4	4
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0	0	0
$\lambda_{X_2}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$\lambda_{S_2}^{(1)} m_2^2$	6	6	6
$\mu_{\text{T3GeneralS1S2}}(\lambda_{10})^*$	0	6	0
$\lambda_{10}(\mu_{\text{T3GeneralS1S2}})^*$	0	6	0
$\mu_{\text{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	12	0	0
$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	12	0	0
$g_2^2 m_2^2$	0	0	0
$g_1^2 m_2^2$	-6	-24	0
$\lambda_{12}^{(1)} m_1^2$	6	6	6
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	0	0	0
$\mu_{\text{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0	0	12
$\lambda_{X_2}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	12
$\lambda_{S_2}^{(1)} m_2^2$	4	4	4
$\mu_{\text{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$\mu_{\text{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0	0	0
$\lambda_{X_5}^{(2)}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	0	0	0

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$\mu_{\text{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0	0	0
$\lambda_{X_5}^{(1)}(\mu_{\text{T3GeneralS1Sq}})^*$	12	0	0
$g_2^2 m_2^2$	-12	-12	-12

$g_1^2 m_2^2$	-6	-24	0
$\lambda_{12}^{(1)} m_1^2$	0	0	0
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	1	1	1
$\mu_{\mathrm{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0	0	20
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0	0	20
$\lambda_{S_2}^{(1)} m_2^2$	0	0	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0	0	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	4	0	0
$\lambda_{X_5}^{(2)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	4	0	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1	1	1

$\mathcal{S}$	E ($\alpha = -4$)
$\mu_{\mathrm{T3GeneralS1Sq}}(\lambda_{X_5}^{(1)})^*$	0
$\lambda_{X_5}^{(1)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0
$g_2^2 m_2^2$	-12
$g_1^2 m_2^2$	-6
$\lambda_{12}^{(1)} m_1^2$	0
$m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	1
$\mu_{\mathrm{T3GeneralS2Sq}}(\lambda_{X_2})^*$	0
$\lambda_{X_2}(\mu_{\mathrm{T3GeneralS2Sq}})^*$	0
$\lambda_{S_2}^{(1)} m_2^2$	0
$\mu_{\mathrm{T3GeneralS1S2}}(\lambda_{10})^*$	0
$\lambda_{10}(\mu_{\mathrm{T3GeneralS1S2}})^*$	0
$\mu_{\mathrm{T3GeneralS1Sq}}(3(\lambda_{X_5}^{(1)})^* + (\lambda_{X_5}^{(2)})^*)$	0
$\lambda_{X_5}^{(2)}(\mu_{\mathrm{T3GeneralS1Sq}})^*$	0
$m_1^2(6\lambda_{12}^{(1)} + 4\lambda_{12}^{(\times)})$	1

$\beta^{(1)}_{\mu_{\mathrm{T3GeneralS1S2}}}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$\lambda_9(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	0	0
$g_2^2 \mu_{\mathrm{T3GeneralS1S2}}$	0	0	0
$g_1^2 \mu_{\mathrm{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)} \mu_{\mathrm{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)} \mu_{\mathrm{T3GeneralS1S2}}$	0	0	0
$\lambda_{10} m_2^2$	0	0	0
$\lambda_{11} m_1^2$	0	0	0

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$\lambda_9(\mu_{\mathrm{T3GeneralS1S2}})^*$	0	12	0
$g_2^2 \mu_{\mathrm{T3GeneralS1S2}}$	0	$-\frac{9}{2}$	0
$g_1^2 \mu_{\mathrm{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{12}^{(1)} \mu_{\mathrm{T3GeneralS1S2}}$	0	2	0
$\lambda_{12}^{(A)} \mu_{\mathrm{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{10} m_2^2$	0	6	0
$\lambda_{11} m_1^2$	0	6	0

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)

$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	12	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{9}{2}$	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	2	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	$-\frac{3}{2}$	0
$\lambda_{10}m_2^2$	0	6	0
$\lambda_{11}m_1^2$	0	6	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{10}m_2^2$	0	0	0
$\lambda_{11}m_1^2$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0	0	0
$\lambda_{10}m_2^2$	0	0	0
$\lambda_{11}m_1^2$	0	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_9(\mu_{\text{T3GeneralS1S2}})^*$	0
$g_2^2\mu_{\text{T3GeneralS1S2}}$	0
$g_1^2\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{12}^{(1)}\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{12}^{(A)}\mu_{\text{T3GeneralS1S2}}$	0
$\lambda_{10}m_2^2$	0
$\lambda_{11}m_1^2$	0

$\beta^{(1)}_{\mu_{\text{T3GeneralS1Sq}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	24	0	0
$\lambda_{S_1}^{(1)}\mu_{\text{T3GeneralS1Sq}}$	2	0	0
$\lambda_{X_5}^{(1)}m_2^2$	6	0	0
$\lambda_{X_9}m_1^2$	6	0	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$\lambda_{S_1}^{(A)}\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda_{X_5}^{(1)}+2\lambda_{X_5}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)

$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda^{(1)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(1)}_{X_5}m_2^2$	0	0	0
$\lambda_{X_9}m_1^2$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(A)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	0	0	0
$\lambda^{(1)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(1)}_{X_5}m_2^2$	0	0	0
$\lambda_{X_9}m_1^2$	0	0	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$\lambda^{(A)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0	0	0
$m_2^2(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	40	0	0
$\lambda^{(1)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	2	0	0
$\lambda^{(1)}_{X_5}m_2^2$	2	0	0
$\lambda_{X_9}m_1^2$	10	0	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	-12	0	0
$\lambda^{(A)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	-4	0	0
$m_2^2(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	40	0	0
$\lambda^{(1)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	2	0	0
$\lambda^{(1)}_{X_5}m_2^2$	0	0	0
$\lambda_{X_9}m_1^2$	10	0	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	-12	0	0
$\lambda^{(A)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	-4	0	0
$m_2^2(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	1	0	0

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_6}(\mu_{\text{T3GeneralS1Sq}})^*$	0
$\lambda^{(1)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0
$\lambda^{(1)}_{X_5}m_2^2$	0
$\lambda_{X_9}m_1^2$	0
$g_2^2\mu_{\text{T3GeneralS1Sq}}$	0
$\lambda^{(A)}_{S_1}\mu_{\text{T3GeneralS1Sq}}$	0
$m_2^2(6\lambda^{(1)}_{X_5}+2\lambda^{(2)}_{X_5})$	0

$\beta^{(1)}_{\mu_{\text{T3GeneralS2Sq}}}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)

$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0	40	0
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0	-12	0
$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0	2	0
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0	-4	0
$\lambda_{X_2}m_2^2$	0	10	0
$\lambda_{X_7}^{(1)}m_1^2$	0	2	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_2}m_2^2$	0	0	0
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	0
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_2}m_2^2$	0	0	0
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	24
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0	0	2
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0	0	0
$\lambda_{X_2}m_2^2$	0	0	6
$\lambda_{X_7}^{(1)}m_1^2$	0	0	6
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	0

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0	0	40
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0	0	-12
$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0	0	2
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0	0	-4
$\lambda_{X_2}m_2^2$	0	0	10
$\lambda_{X_7}^{(1)}m_1^2$	0	0	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0	0	1

$\mathcal{S}$	E ($\alpha=-4$)
$\lambda_{X_1}(\mu_{\text{T3GeneralS2Sq}})^*$	0
$g_2^2\mu_{\text{T3GeneralS2Sq}}$	0

$\lambda_{S_2}^{(1)}\mu_{\text{T3GeneralS2Sq}}$	0
$\lambda_{S_2}^{(A)}\mu_{\text{T3GeneralS2Sq}}$	0
$\lambda_{X_2}m_2^2$	0
$\lambda_{X_7}^{(1)}m_1^2$	0
$m_1^2(6\lambda_{X_7}^{(1)}+2\lambda_{X_7}^{(2)})$	0

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8	-8	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$	$-\frac{5}{12}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta^{(1)}_{Y_e}$

$\mathcal{S}$	A ($\alpha=0$)	A ($\alpha=-2$)	A ($\alpha=-4$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	B ($\alpha=1$)	B ($\alpha=-1$)	B ($\alpha=-3$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	C ($\alpha=1$)	C ($\alpha=-1$)	C ($\alpha=-3$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	D ($\alpha=0$)	D ($\alpha=2$)	D ($\alpha=-2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha=0$)	E ($\alpha=2$)	E ($\alpha=-2$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$	$-\frac{15}{4}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha=-4$)
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta^{(1)}_{Y_u}$

$\mathcal{S}$	A ($\alpha = 0$)	A ($\alpha = -2$)	A ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	B ($\alpha = 1$)	B ($\alpha = -1$)	B ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	C ($\alpha = 1$)	C ($\alpha = -1$)	C ($\alpha = -3$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	D ($\alpha = 0$)	D ($\alpha = 2$)	D ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha = 0$)	E ($\alpha = 2$)	E ($\alpha = -2$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$	$-\frac{3}{2}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$	$-\frac{9}{4}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8	-8	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$	$-\frac{17}{12}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1	1	1

$\mathcal{S}$	E ($\alpha = -4$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_u)_{ij}$	-8

$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1